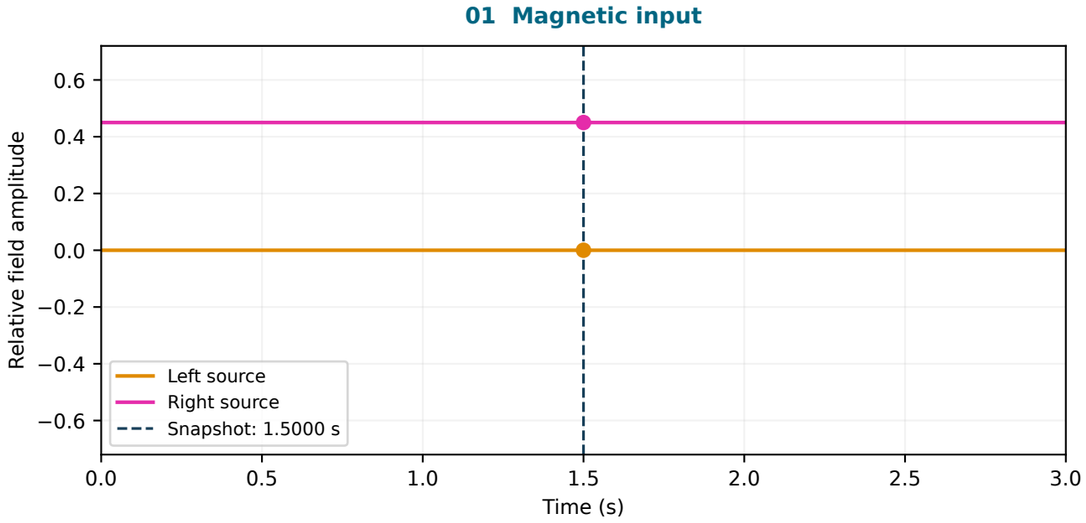
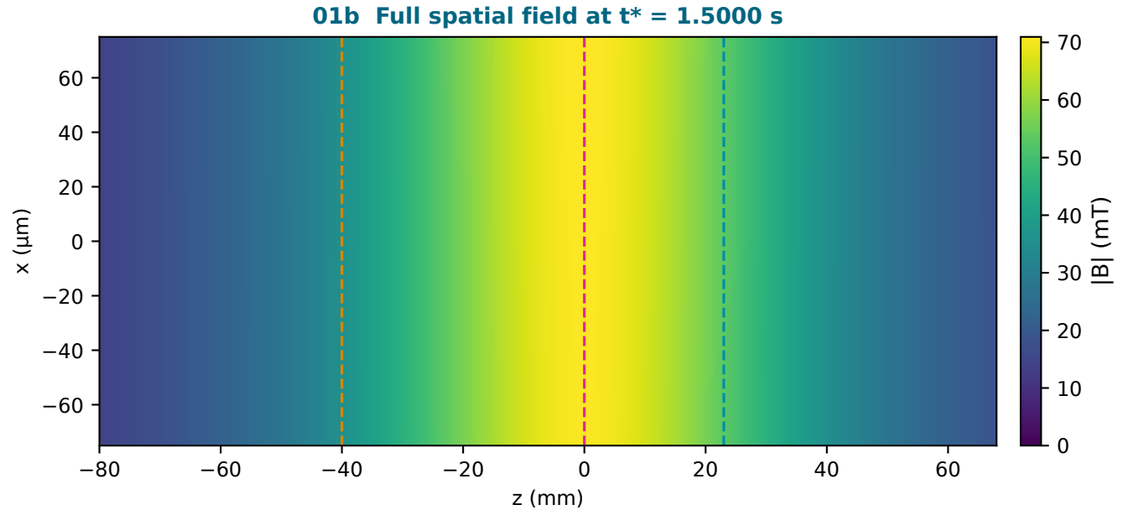
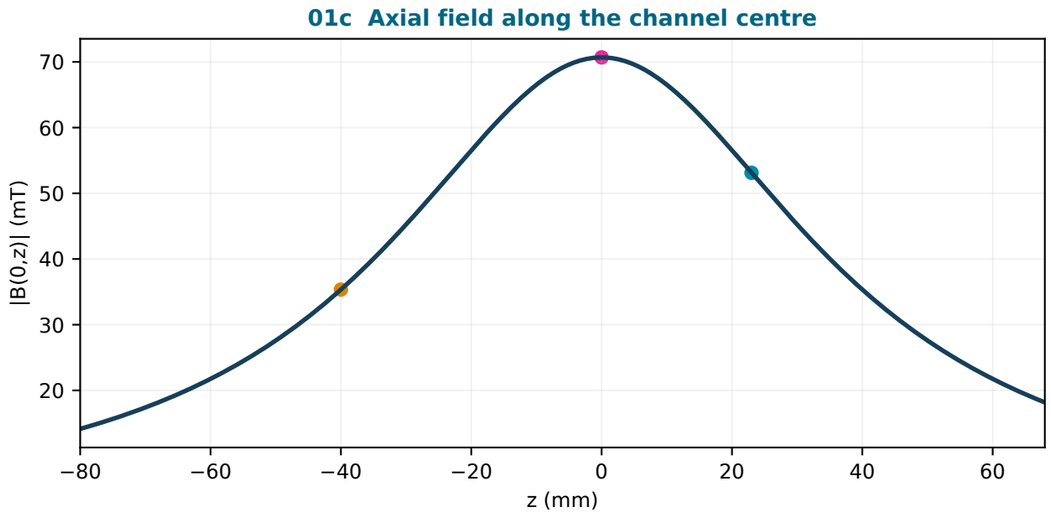
TRAINED MASTER10 | Constant right source

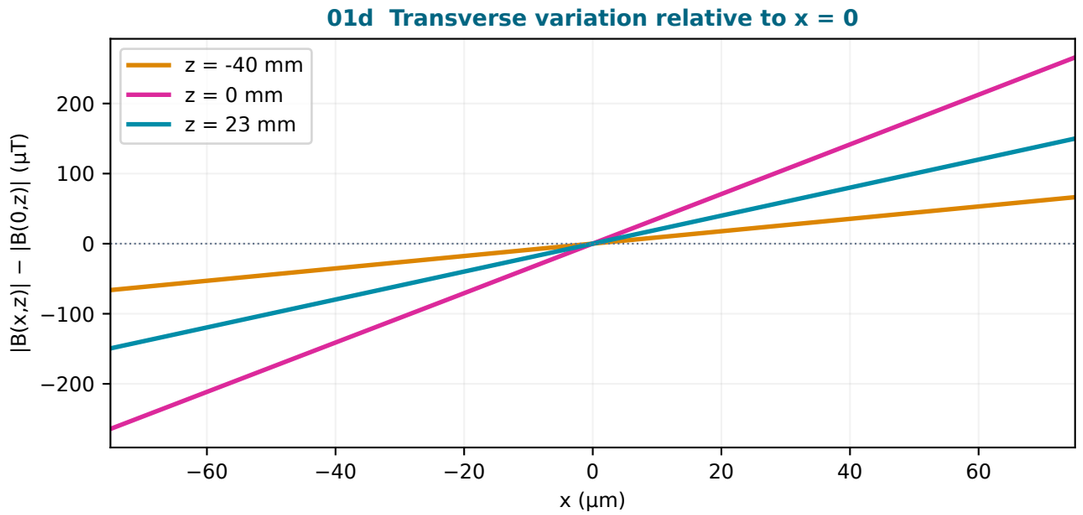
Field detail after input 01 · t*=1.5000 s · a_left=0.0000, a_right=0.4500

Snapshot selection: Constant input: snapshot at midpoint









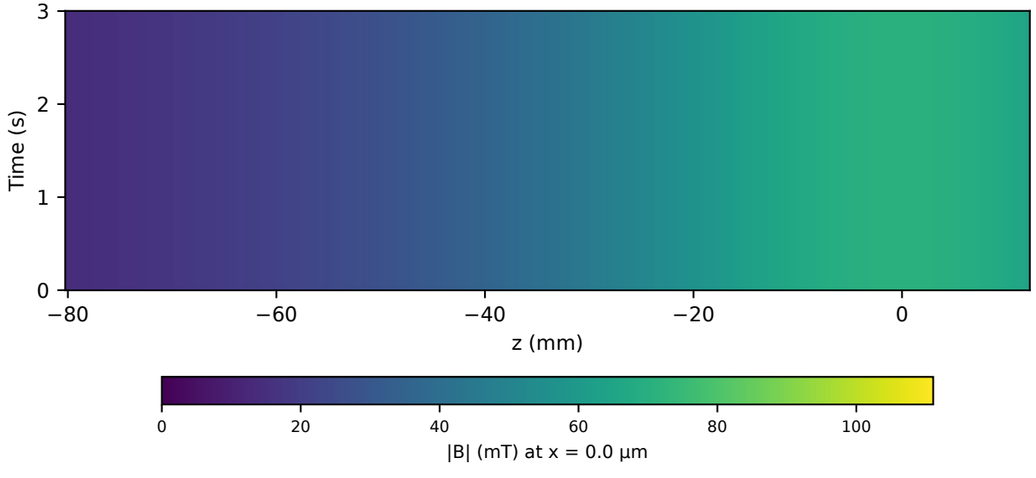
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

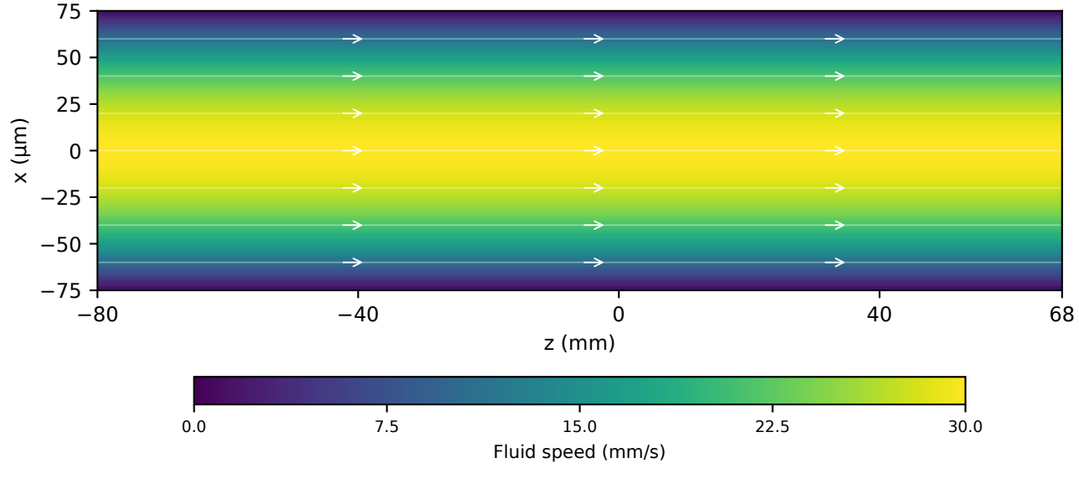
TRAINED MASTER10 | Constant right source

Continuation: panels 02–05 · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s · η=3.277 mPa·s · T=3 s

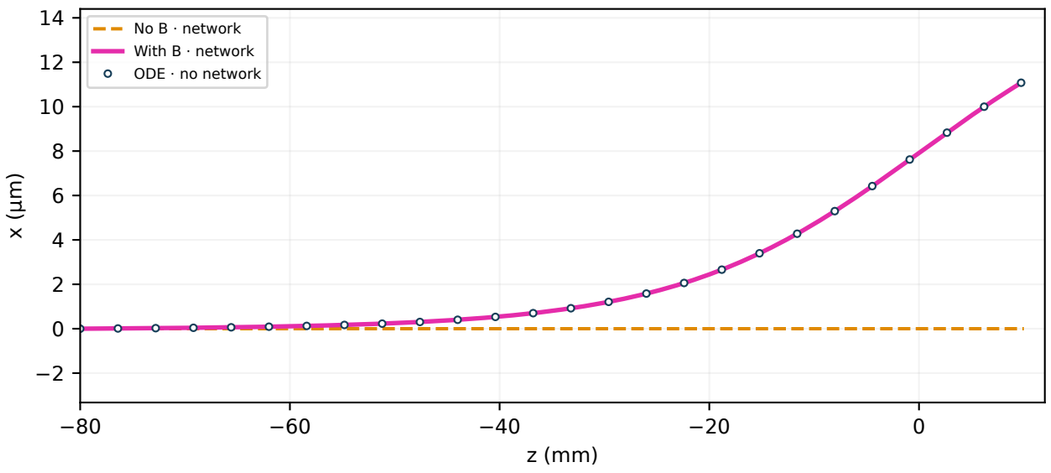
02 Space-time magnetic field



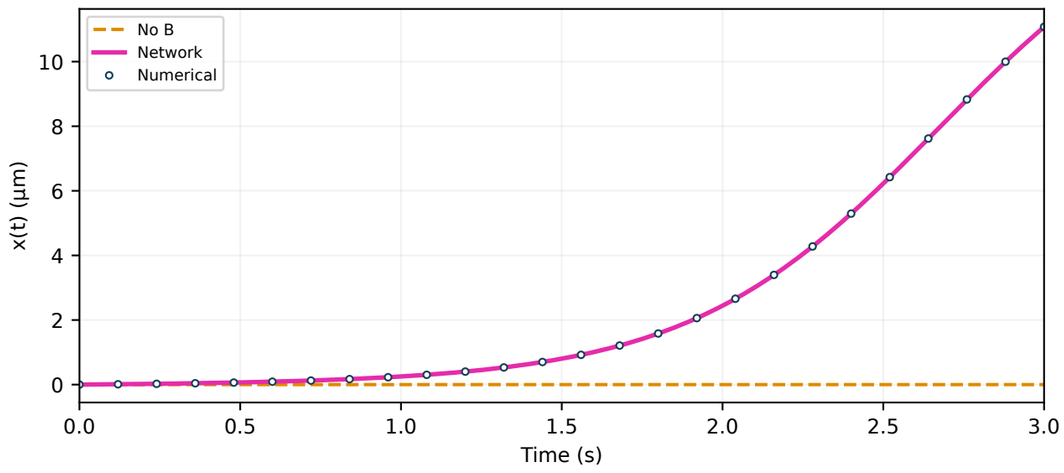
03 Trained-domain fluid flow



04 Network / numerical paths



05 Transverse response



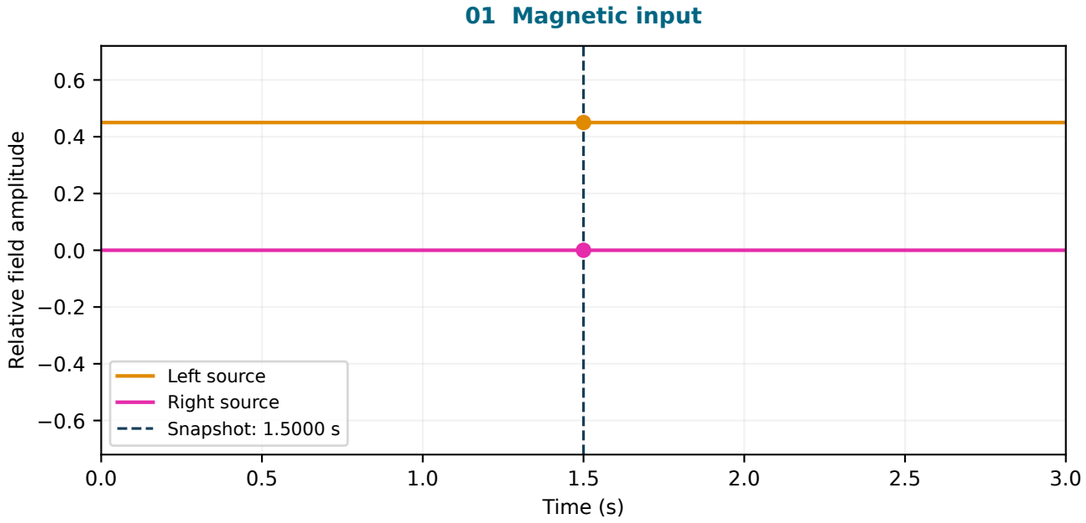
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

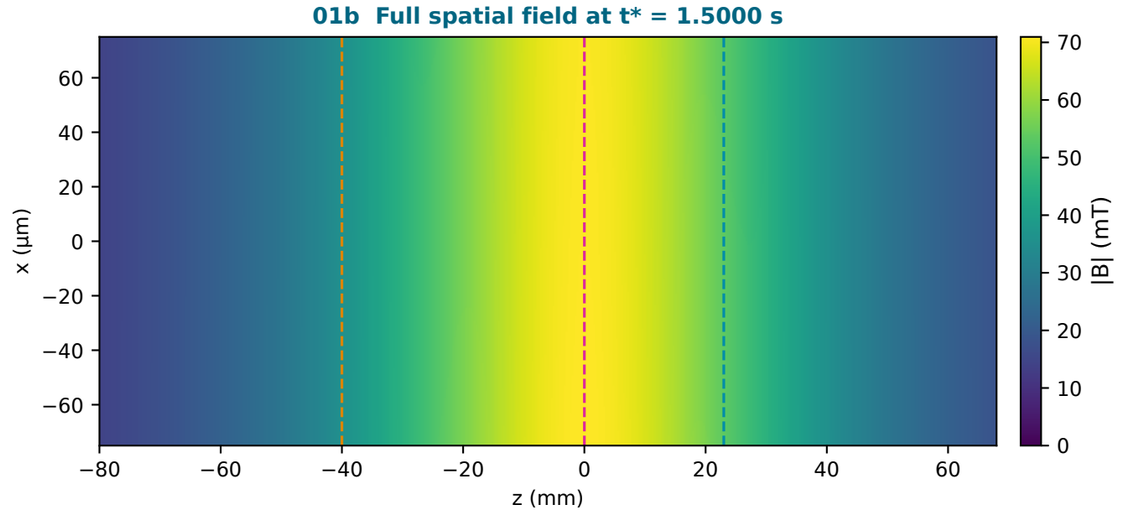
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

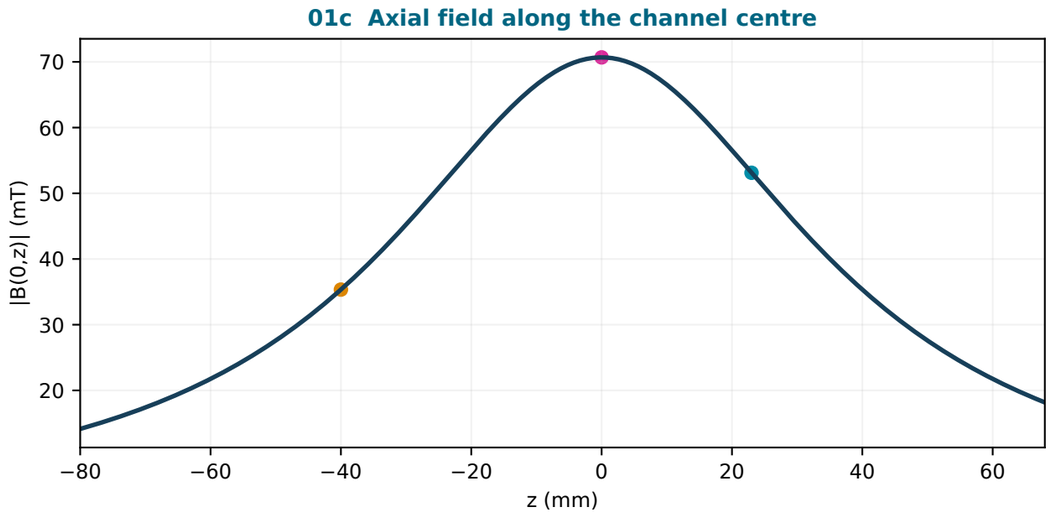
TRAINED MASTER10 | Constant left source

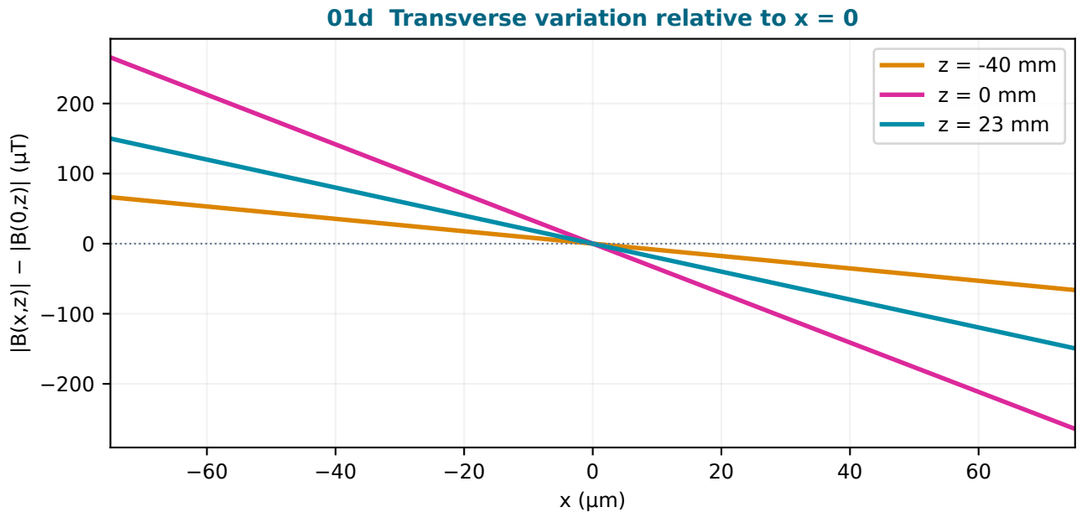
Field detail after input 01 · t*=1.5000 s · a_left=0.4500, a_right=0.0000

Snapshot selection: Constant input: snapshot at midpoint







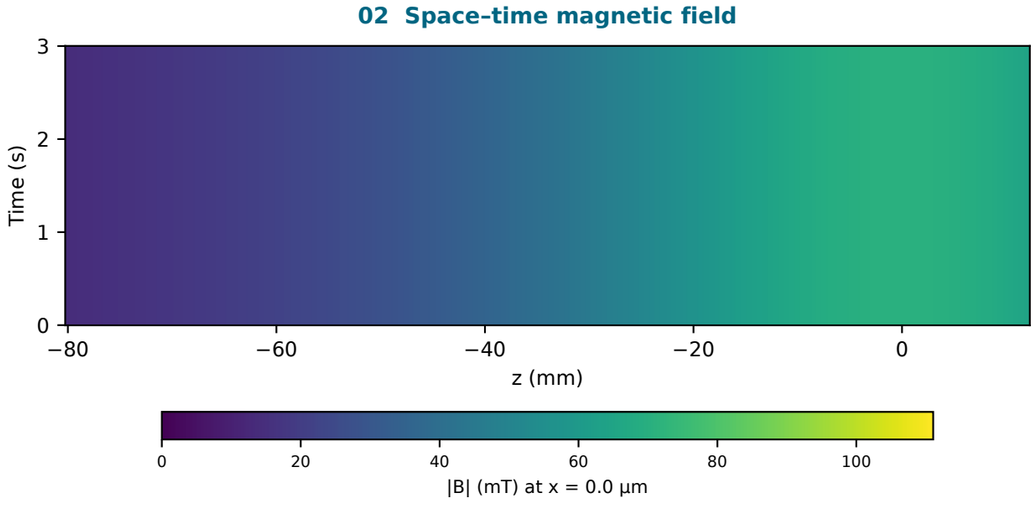


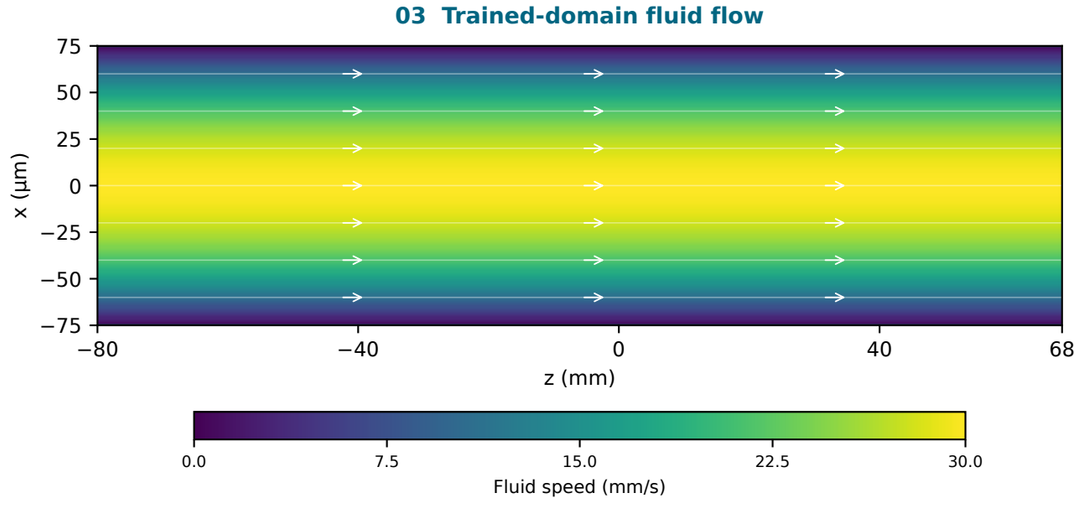
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

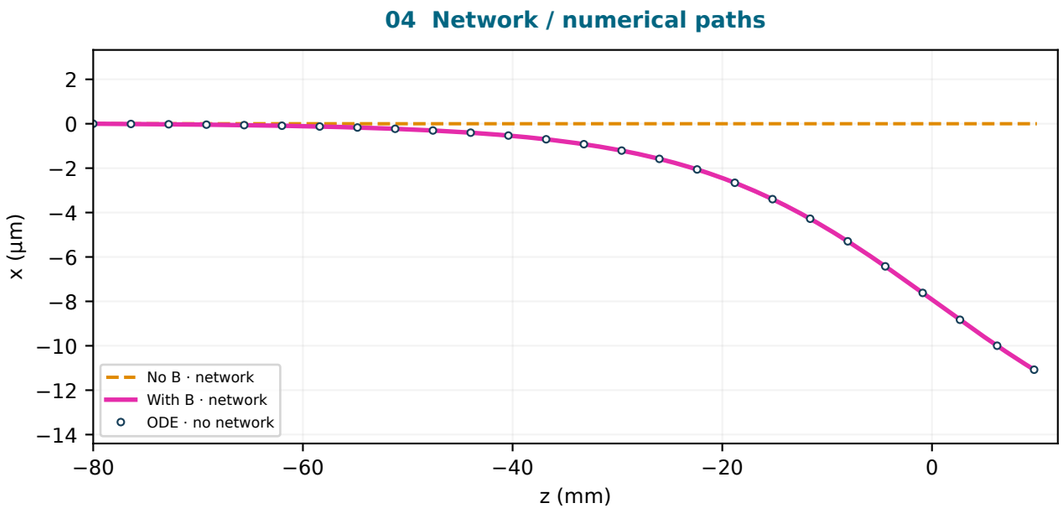
Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

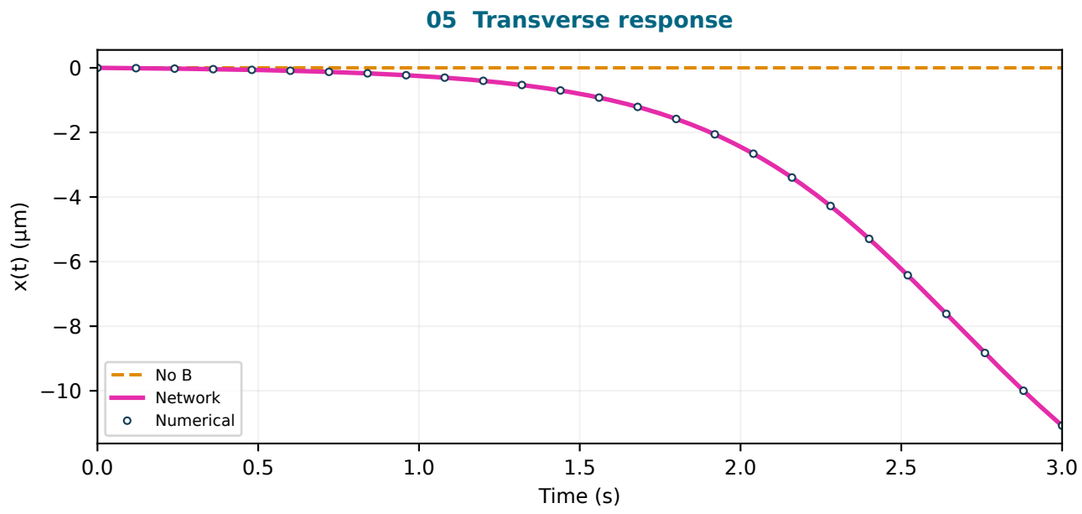
TRAINED MASTER10 | Constant left source

Continuation: panels 02–05 · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s · η=3.277 mPa·s · T=3 s









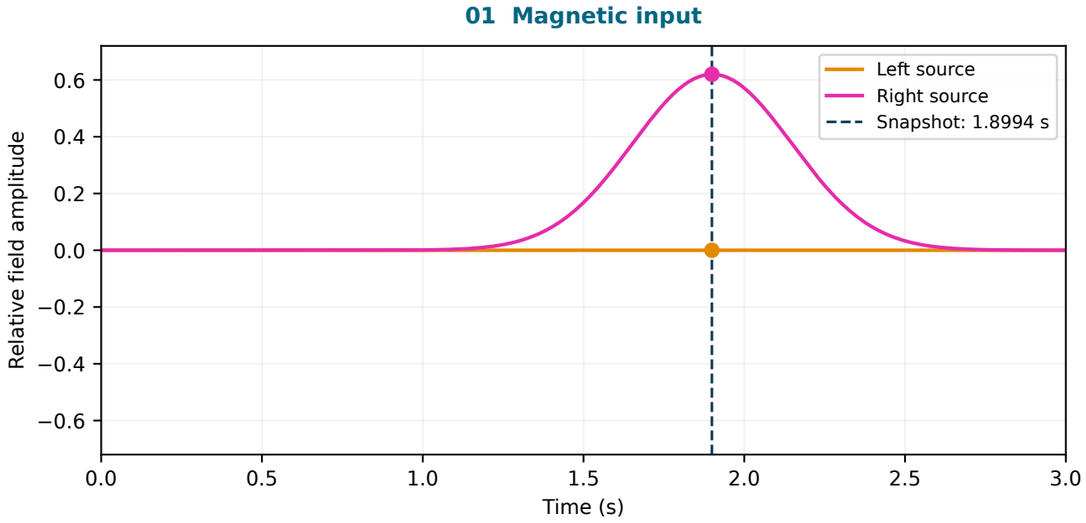
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

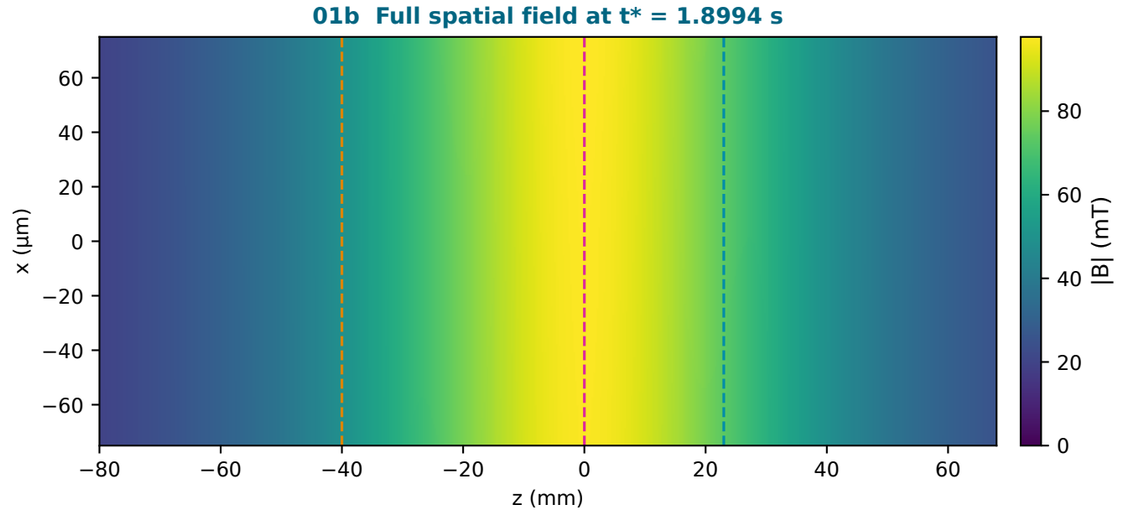
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

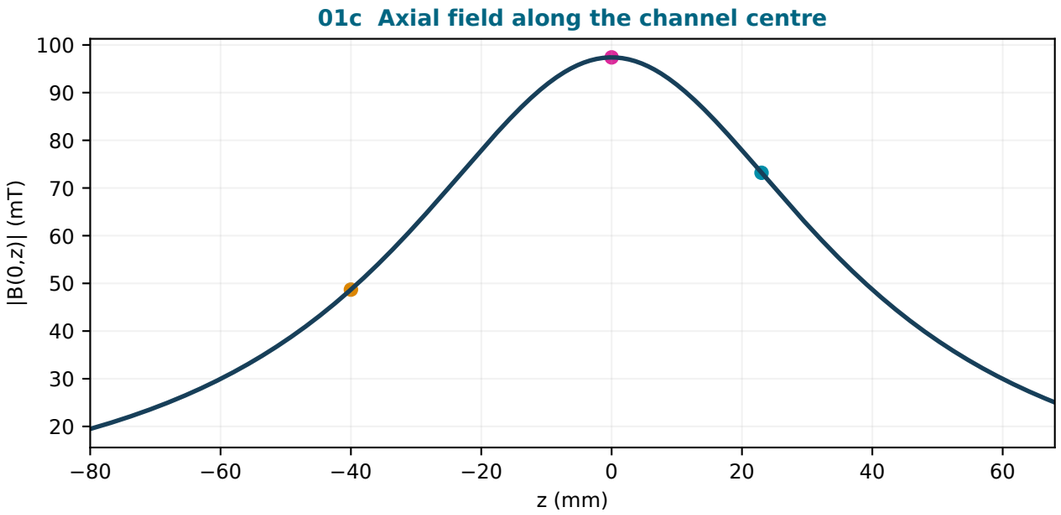
TRAINED MASTER10 | Gaussian pulse on the right

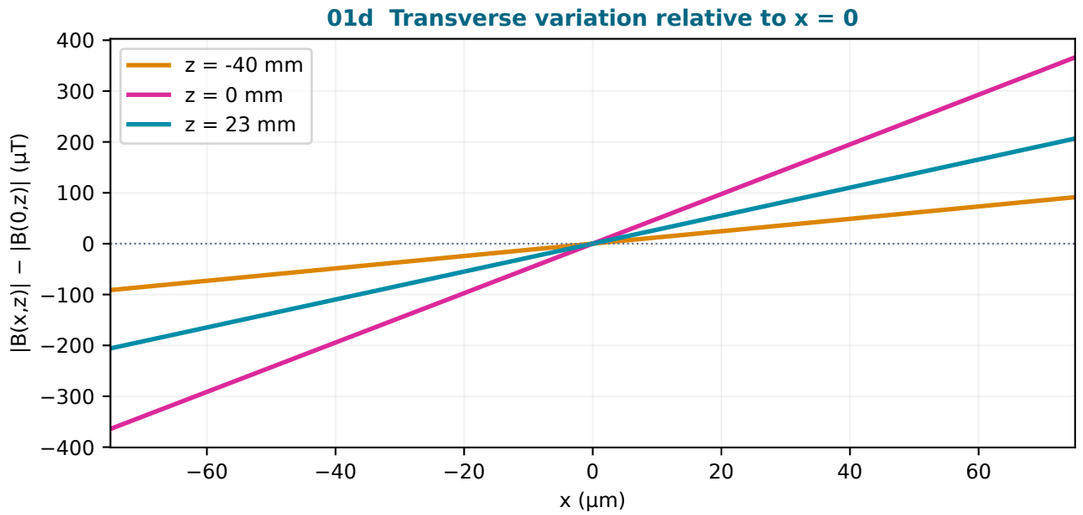
Field detail after input 01 · t*=1.8994 s · a_left=0.0000, a_right=0.6200

Snapshot selection: Peak actual field amplitude









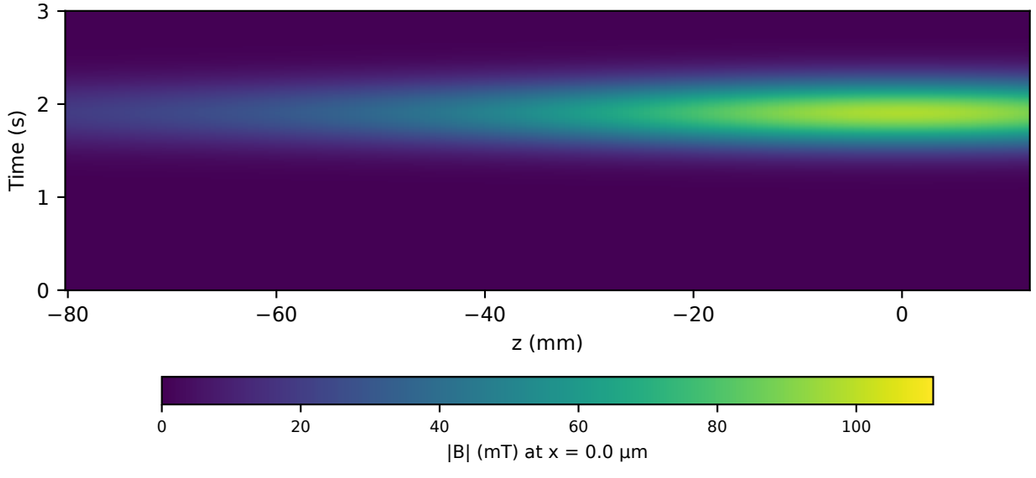
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

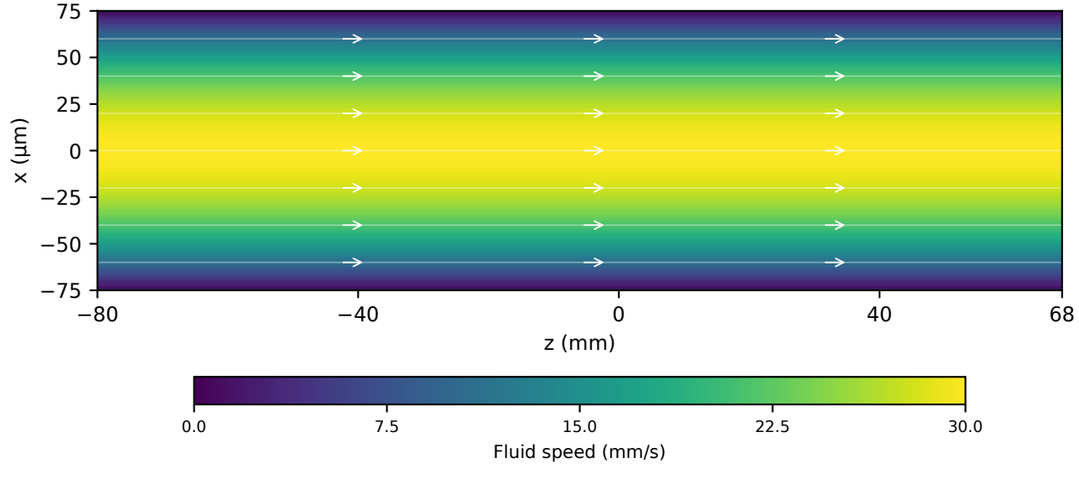
TRAINED MASTER10 | Gaussian pulse on the right

Continuation: panels 02–05 · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s · η=3.277 mPa·s · T=3 s

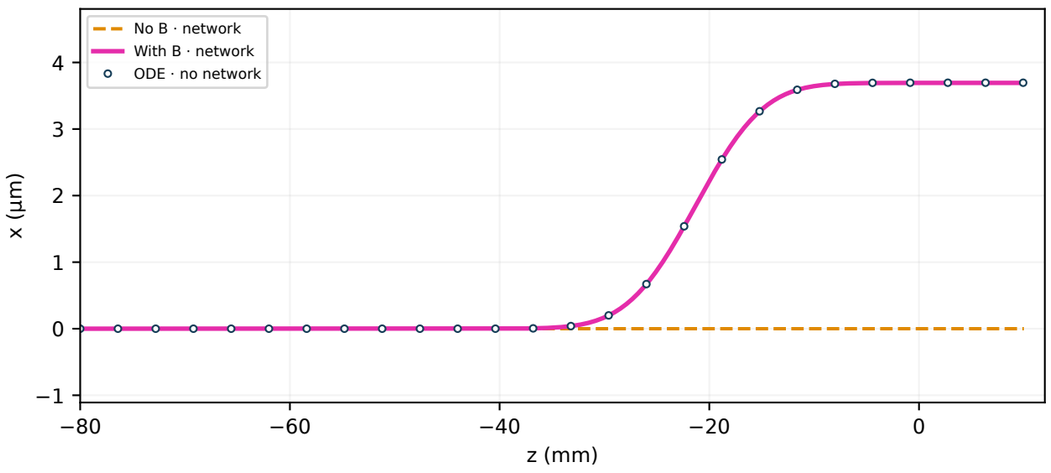
02 Space–time magnetic field



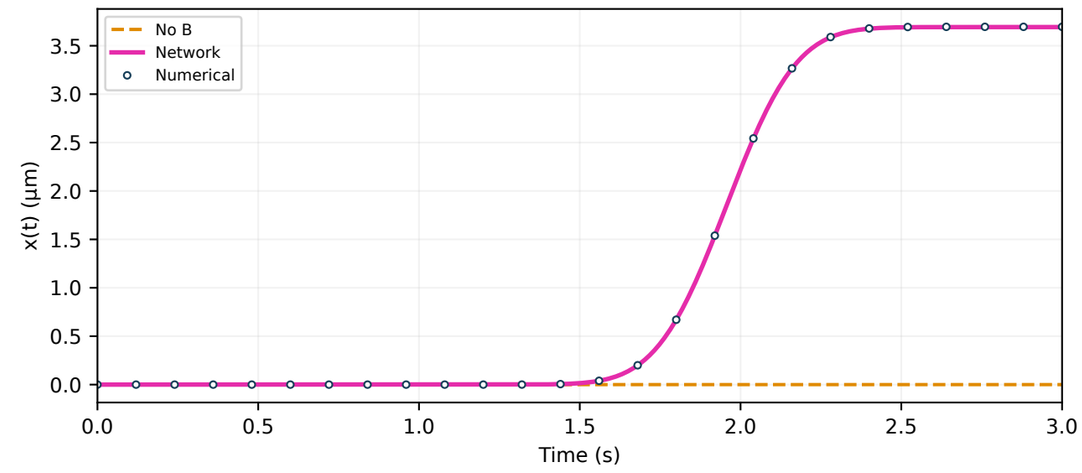
03 Trained-domain fluid flow



04 Network / numerical paths



05 Transverse response



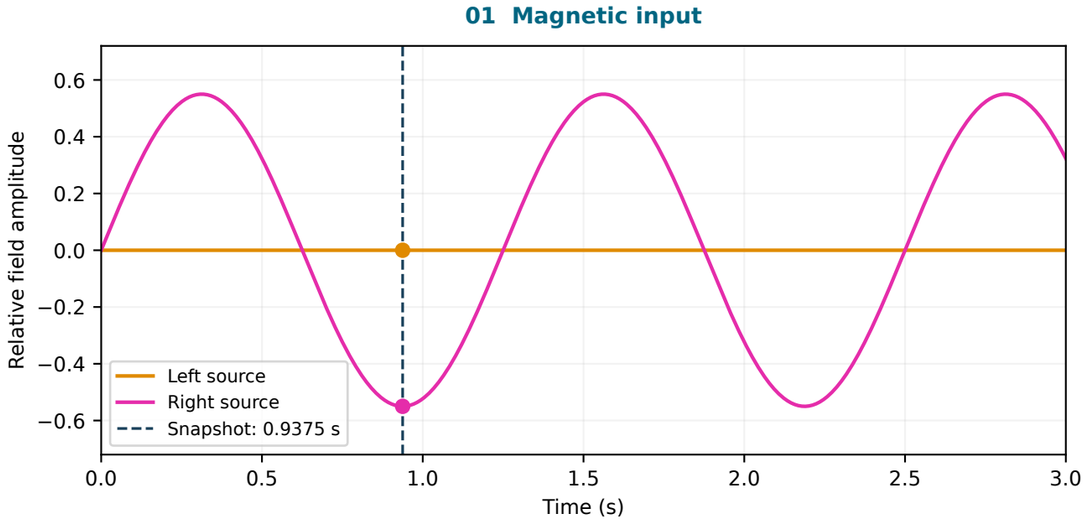
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

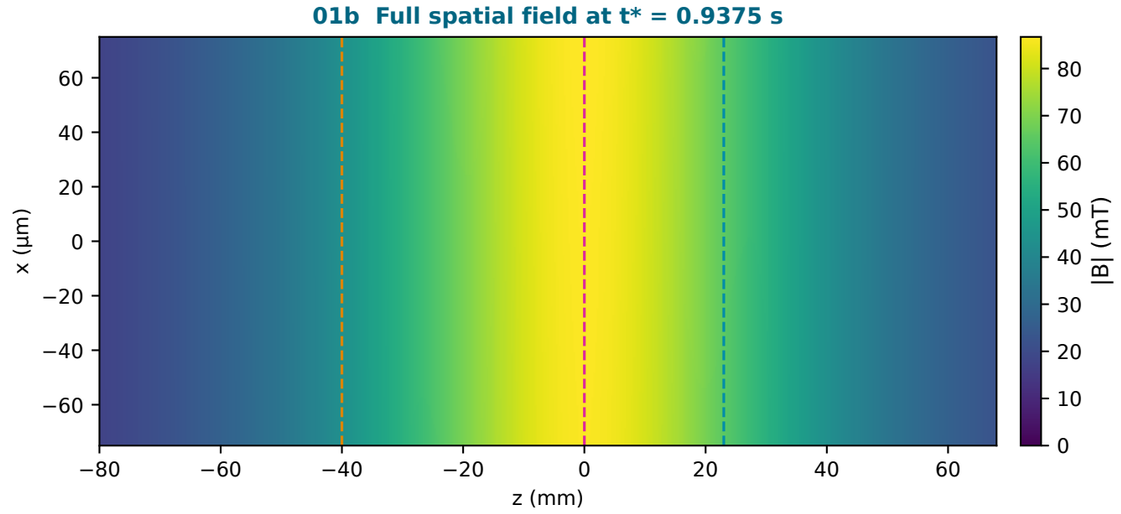
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

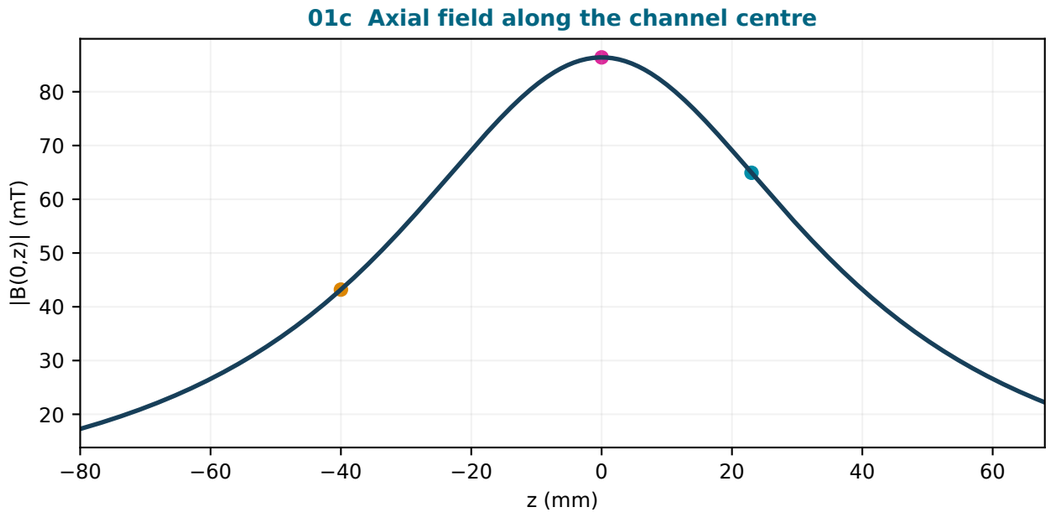
TRAINED MASTER10 | Sinusoidal right source

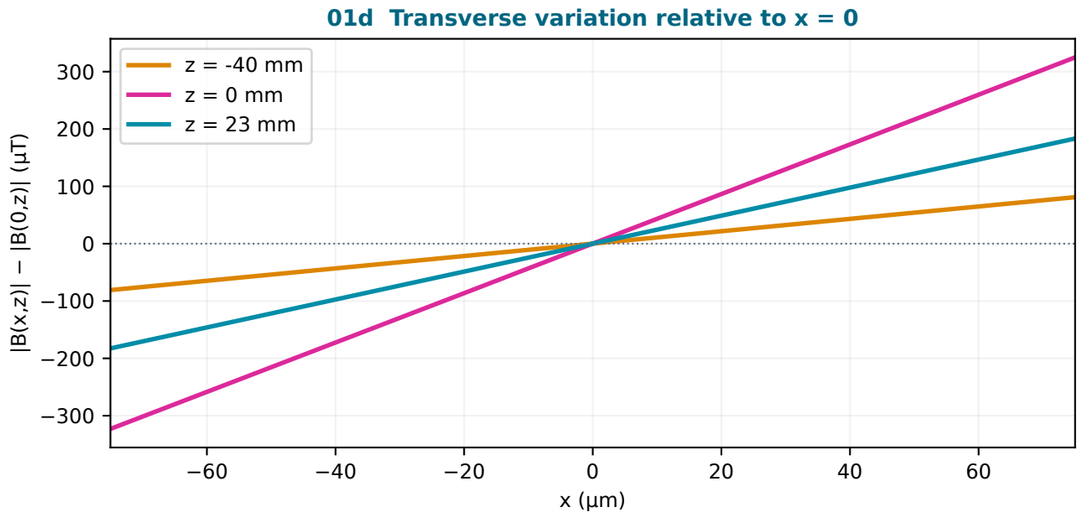
Field detail after input 01 · t*=0.9375 s · a_left=0.0000, a_right=-0.5500

Snapshot selection: Peak actual field amplitude







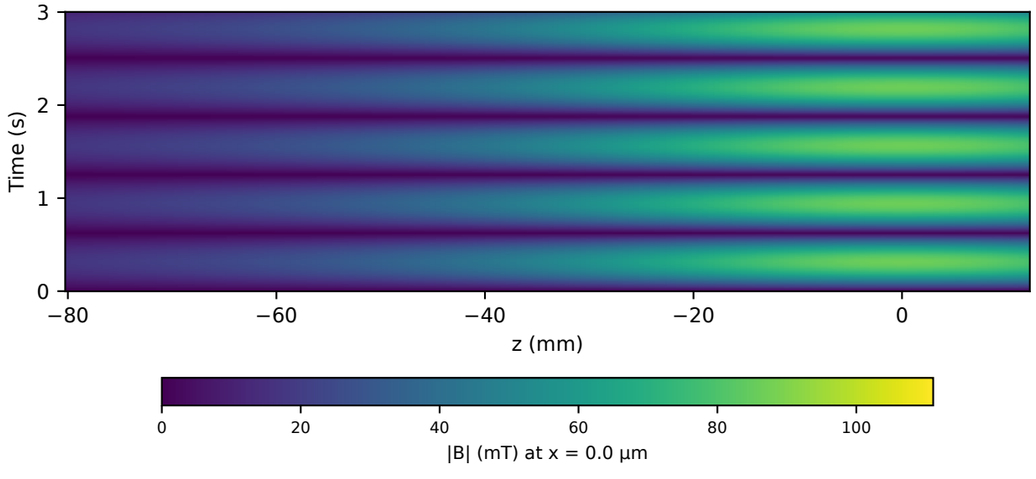


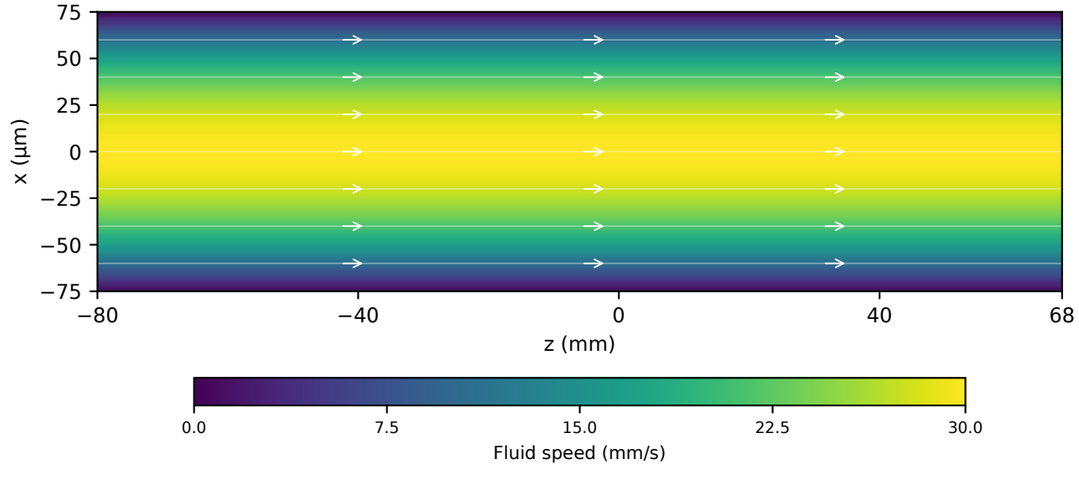
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

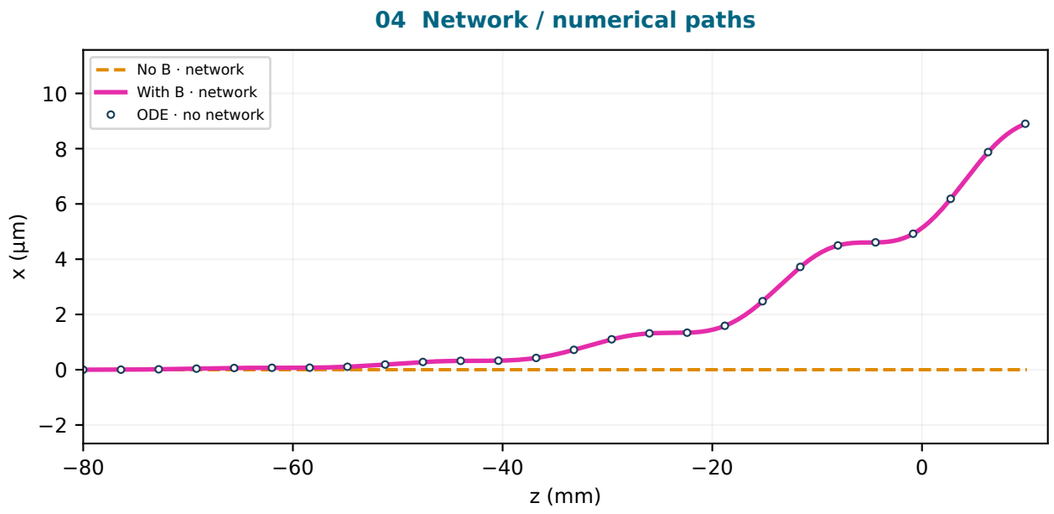
Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

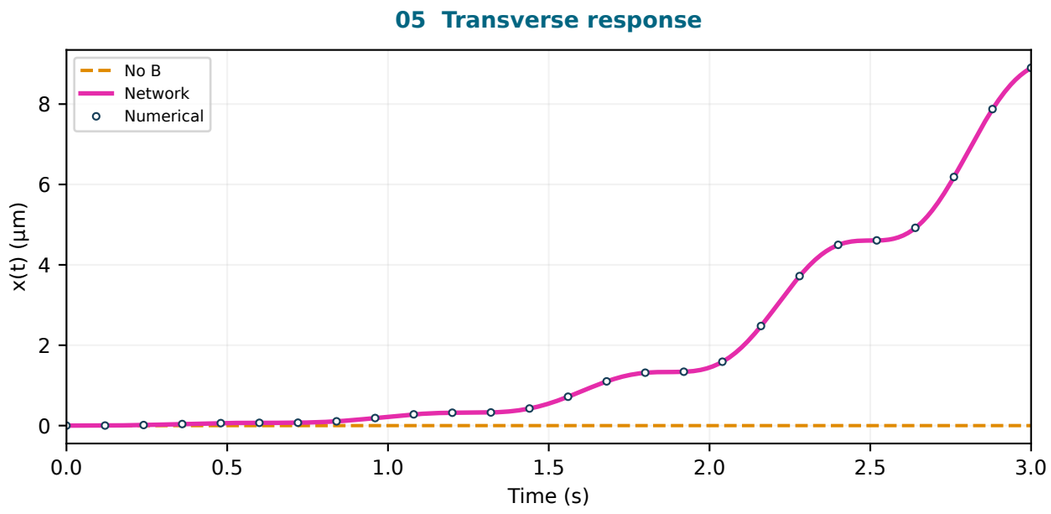
TRAINED MASTER10 | Sinusoidal right source

Continuation: panels 02–05 · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s · η=3.277 mPa·s · T=3 s









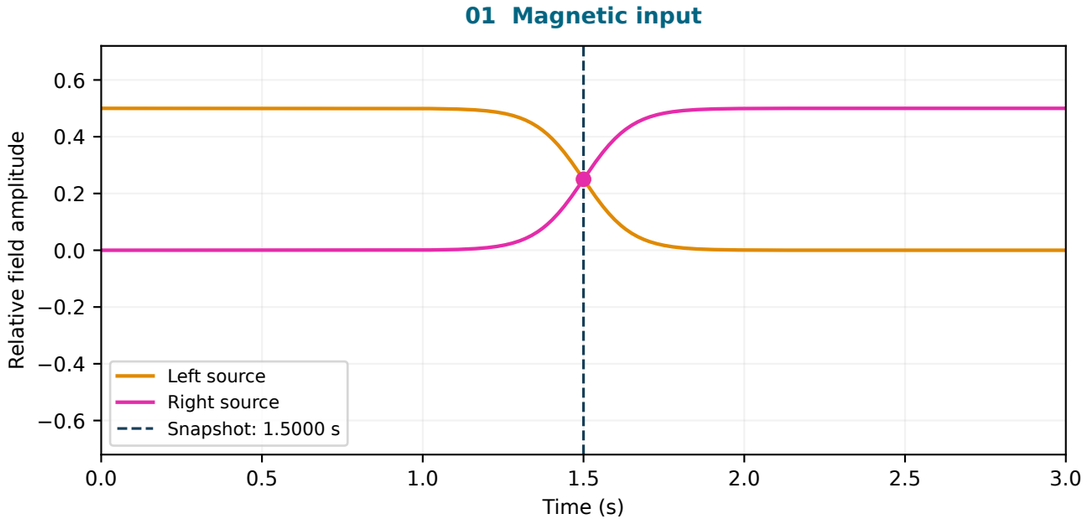
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

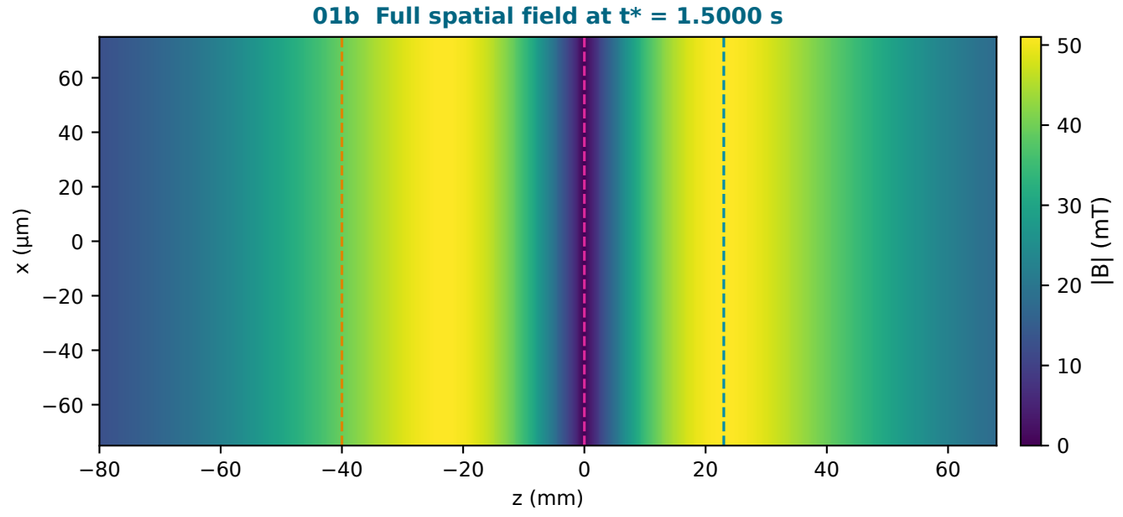
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

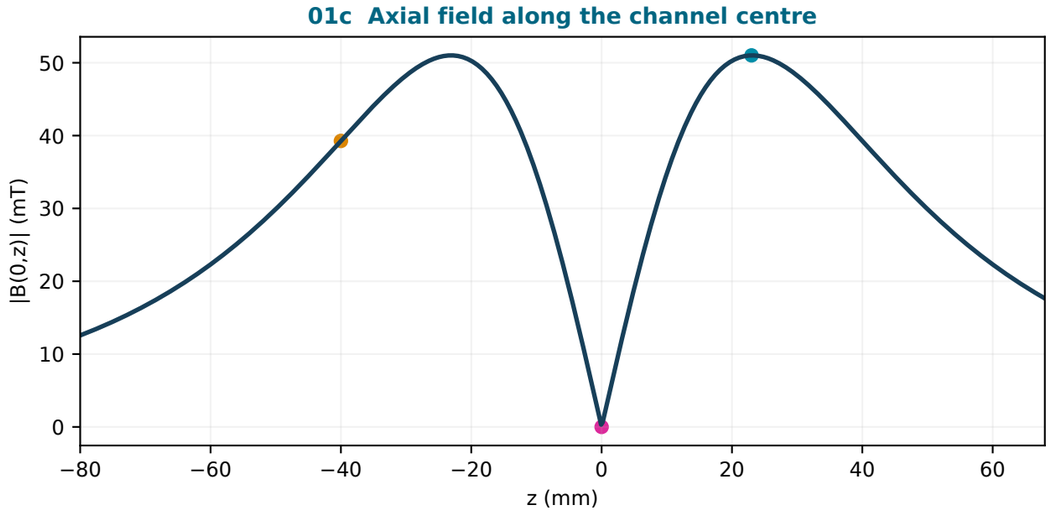
TRAINED MASTER10 | Smooth switch: left to right

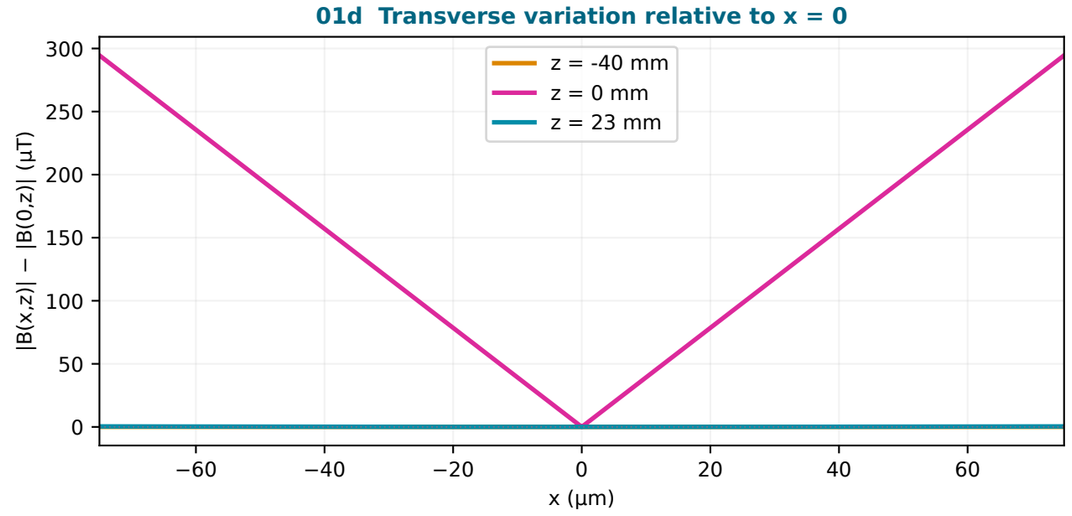
Field detail after input 01 · t*=1.5000 s · a_left=0.2500, a_right=0.2500

Snapshot selection: Strong axial interference notch









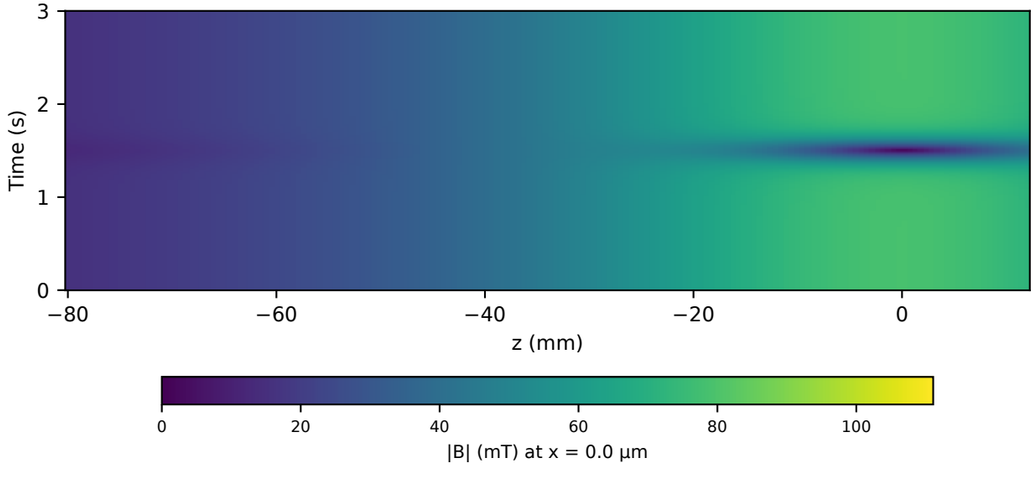
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

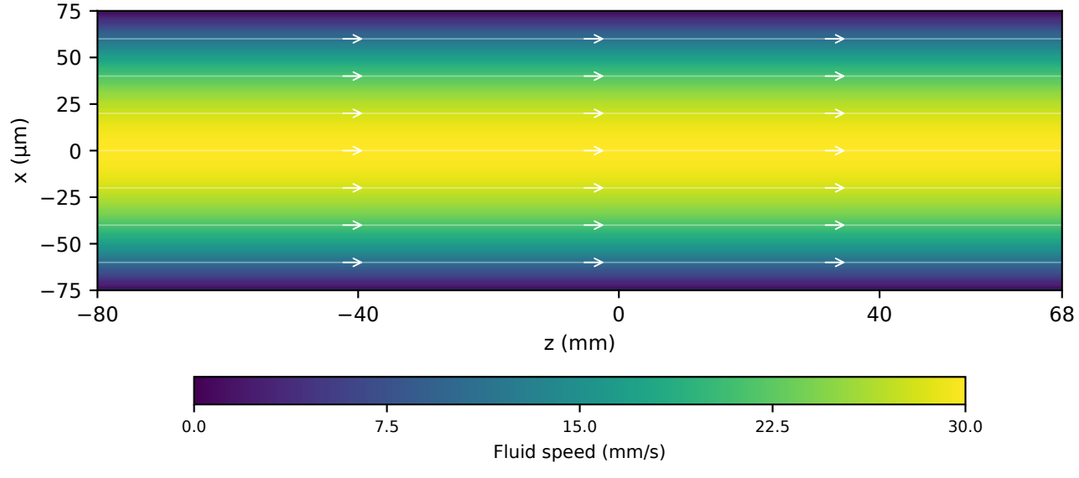
TRAINED MASTER10 | Smooth switch: left to right

Continuation: panels 02–05 · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s · η=3.277 mPa·s · T=3 s

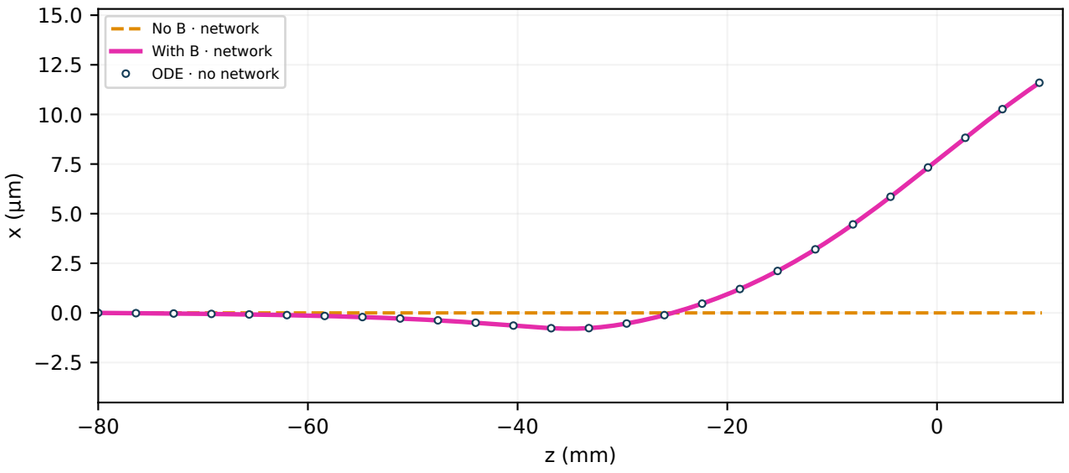
02 Space–time magnetic field



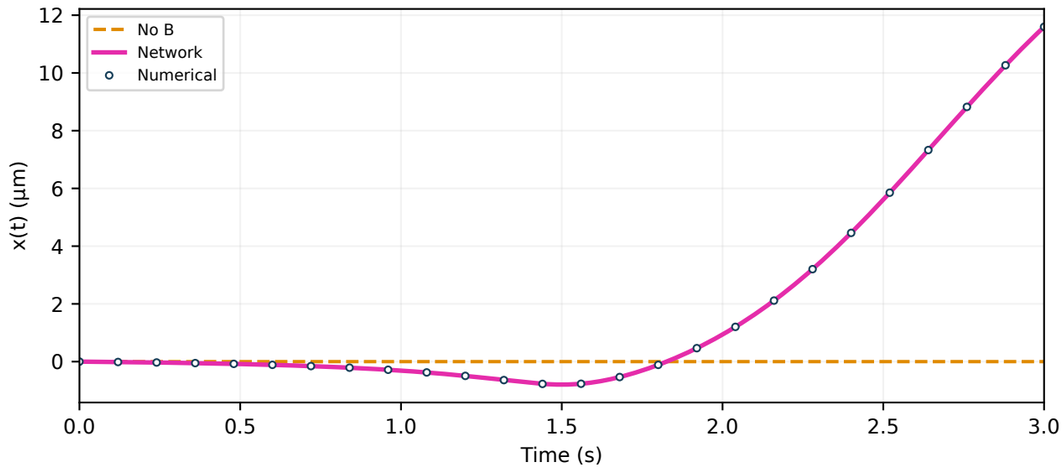
03 Trained-domain fluid flow



04 Network / numerical paths



05 Transverse response



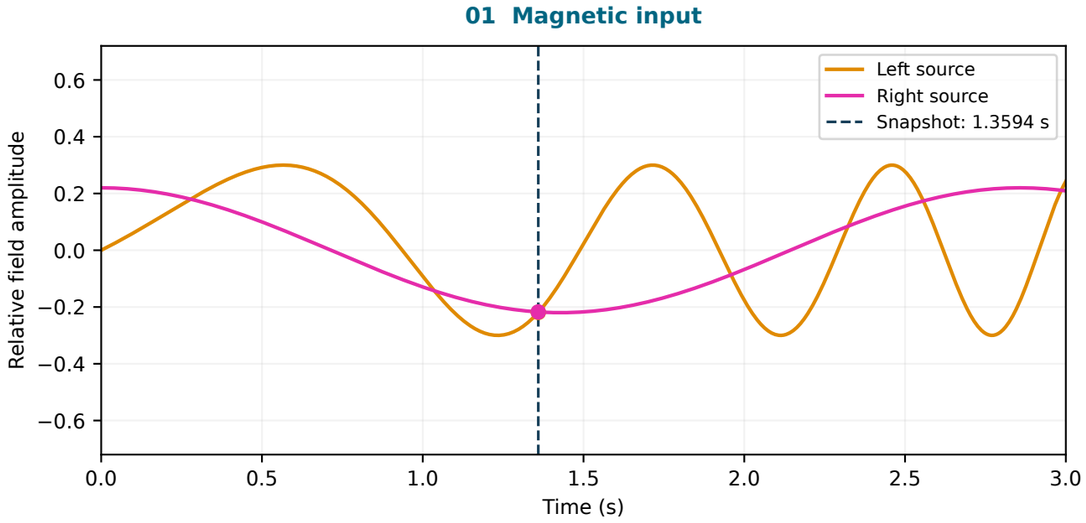
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

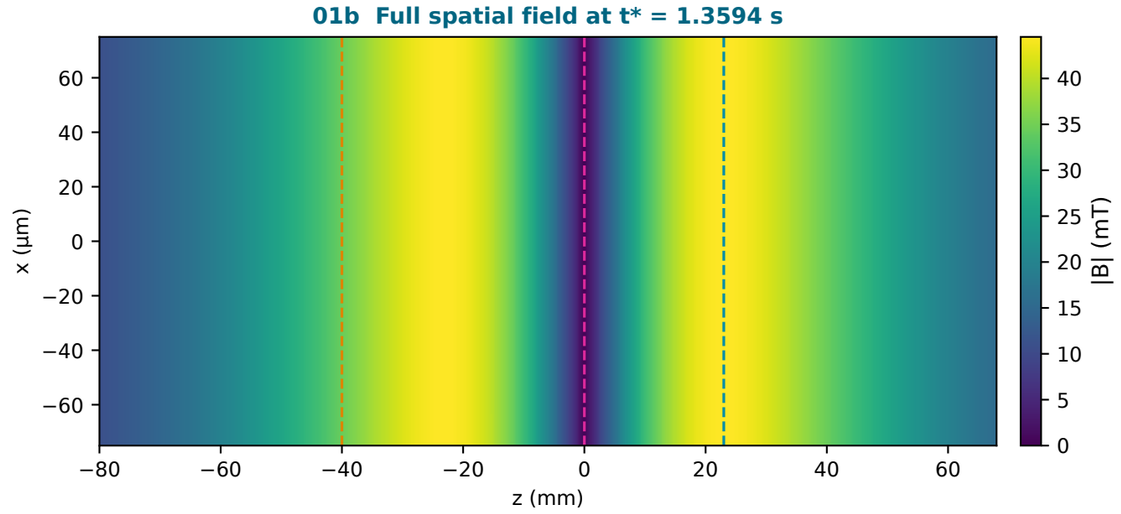
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

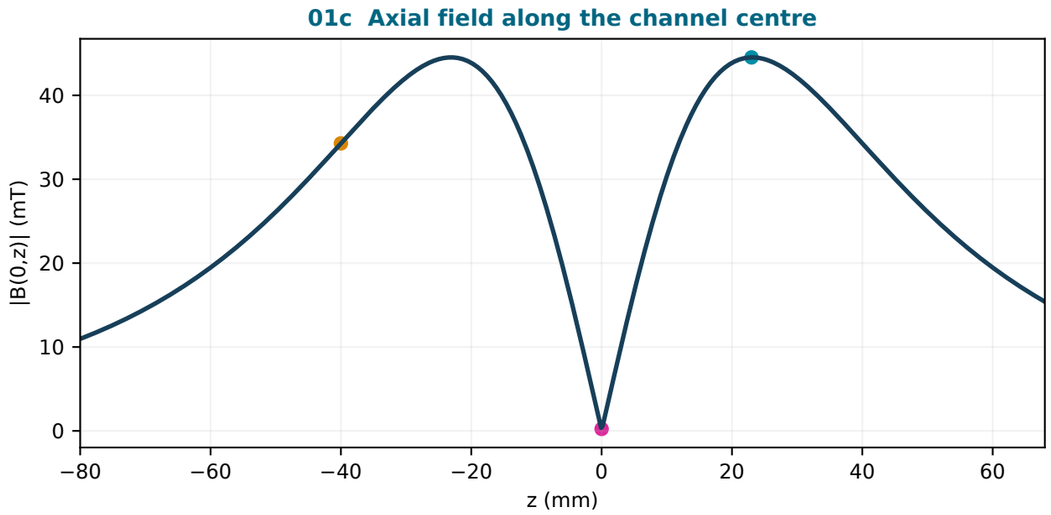
TRAINED MASTER10 | Two-source chirp and cosine

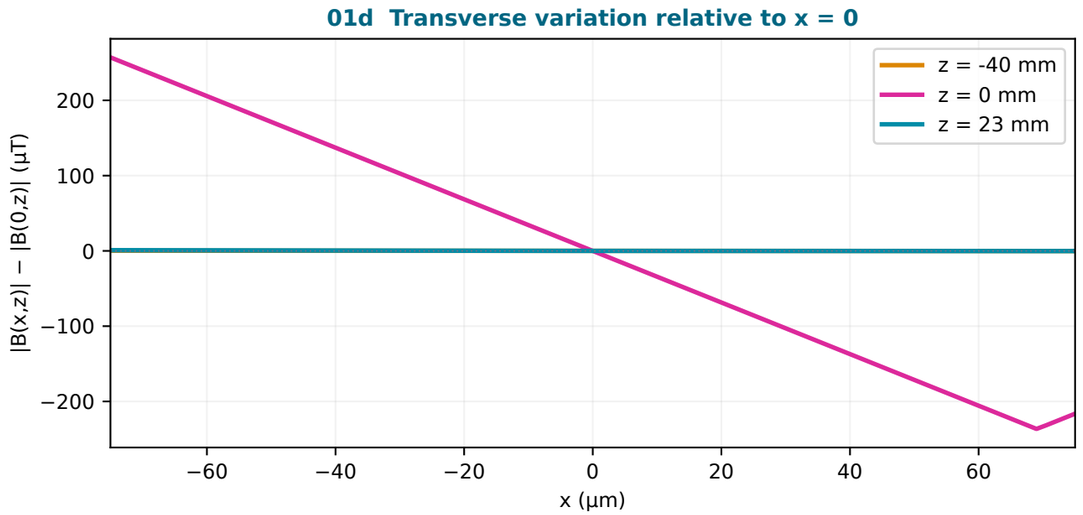
Field detail after input 01 · t*=1.3594 s · a_left=-0.2190, a_right=-0.2175

Snapshot selection: Strong axial interference notch







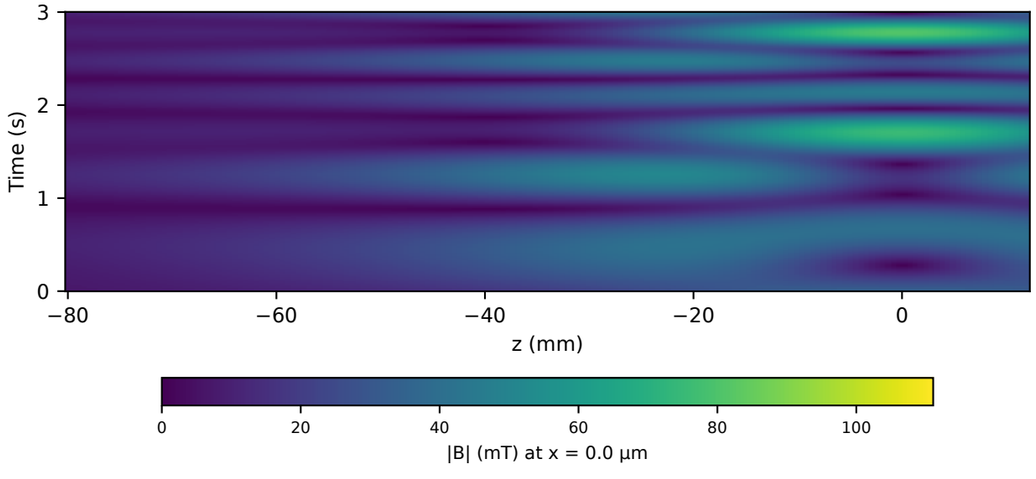


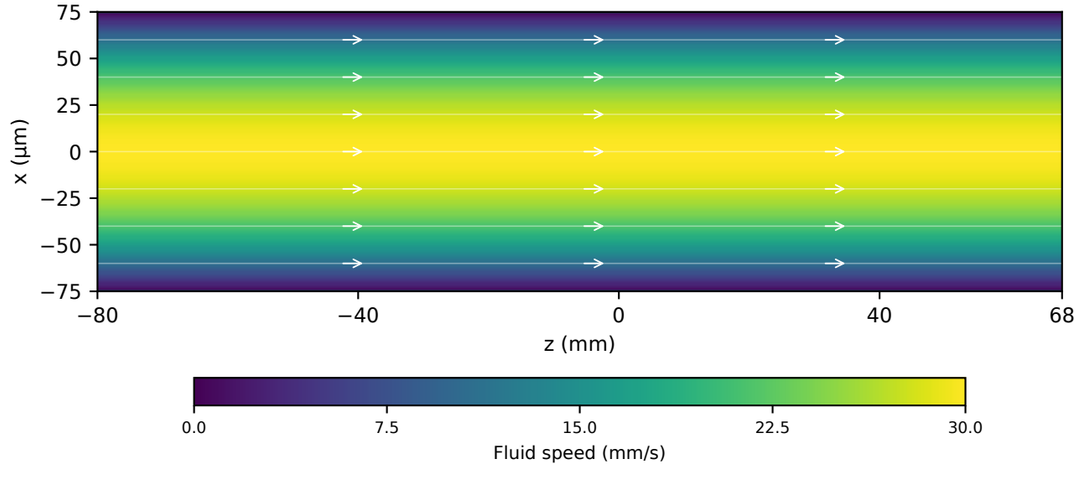
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

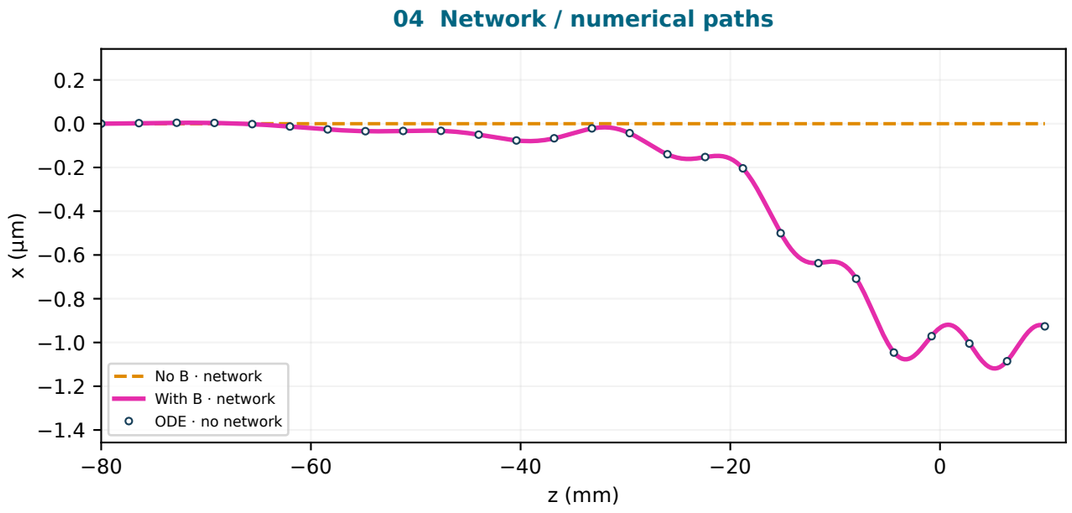
Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

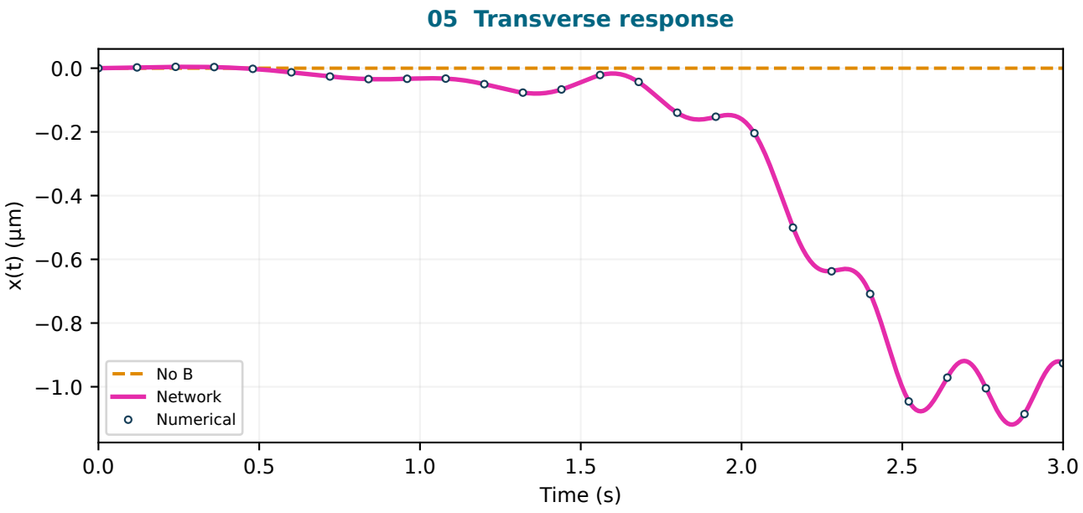
TRAINED MASTER10 | Two-source chirp and cosine

Continuation: panels 02–05 · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s · η=3.277 mPa·s · T=3 s









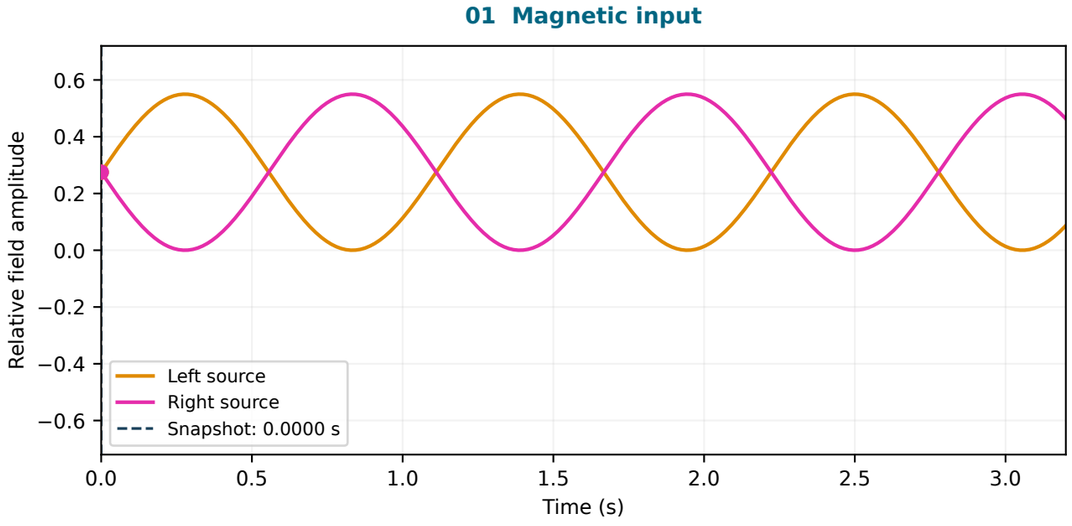
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

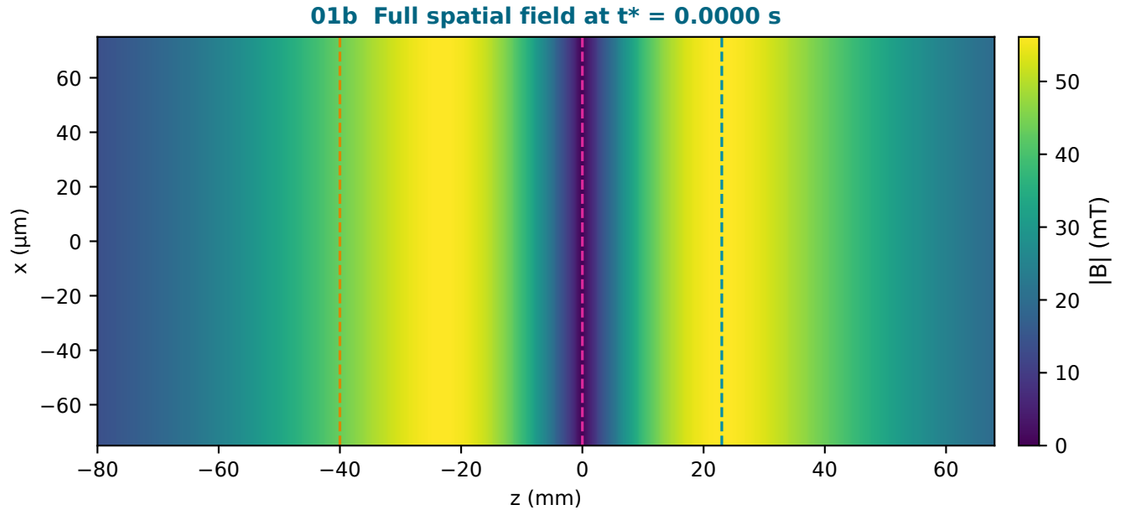
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

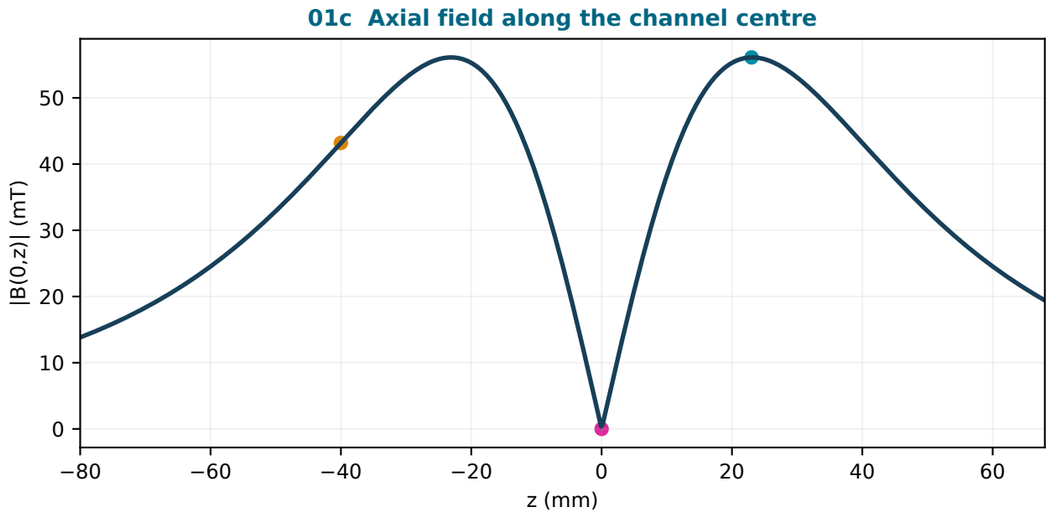
TRAINED MASTER10 | Off-axis release · alternating attraction

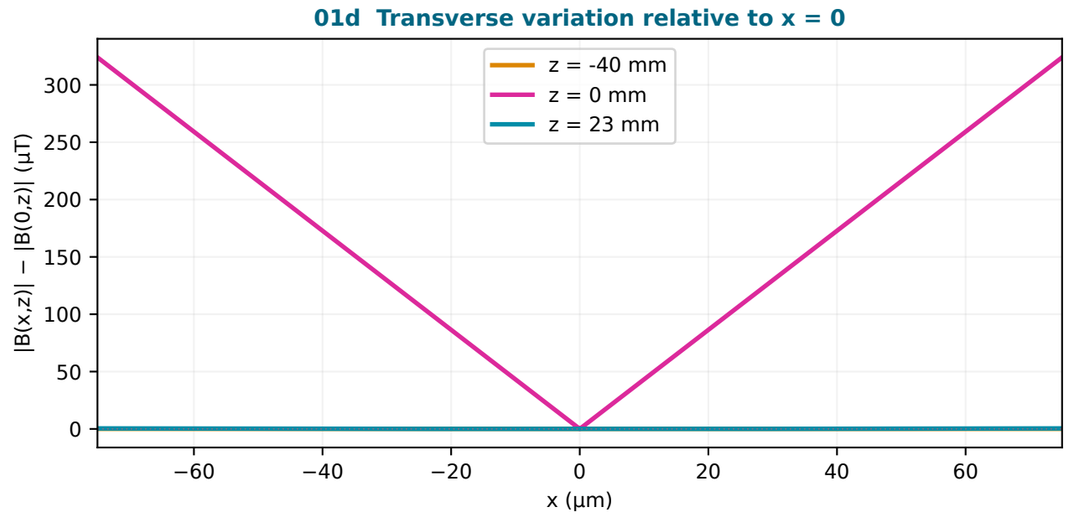
Field detail after input 01 · t*=0.0000 s · a_left=0.2750, a_right=0.2750

Snapshot selection: Strong axial interference notch









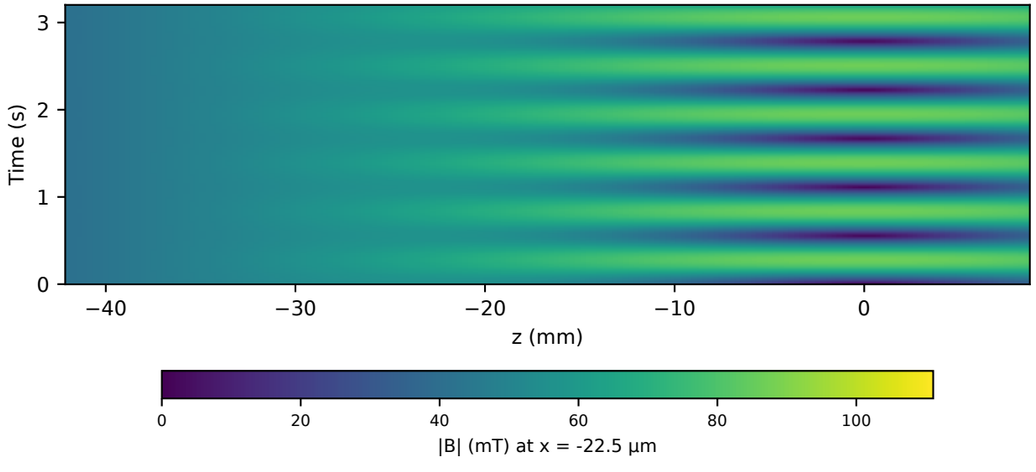
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

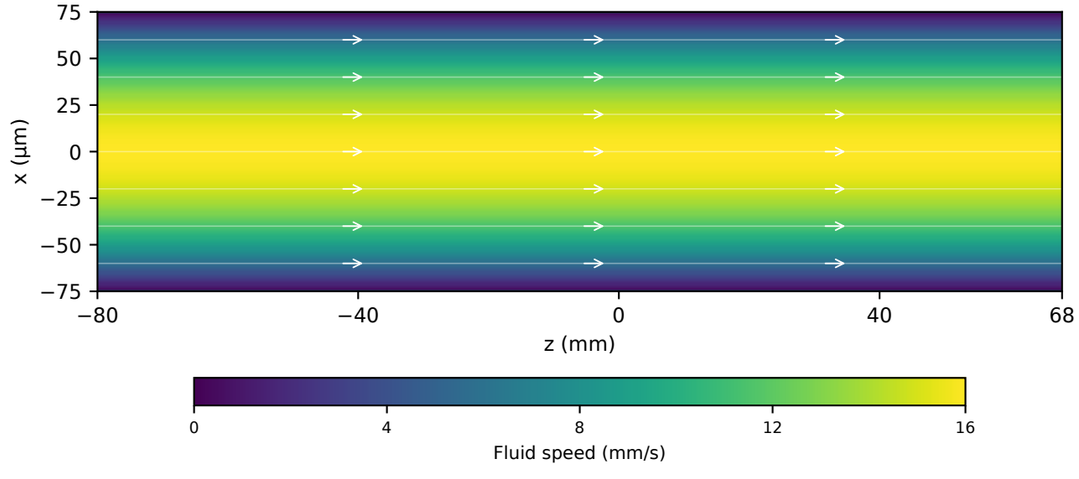
TRAINED MASTER10 | Off-axis release · alternating attraction

Continuation: panels 02–05 · x_0=-22.50 µm, z_0=-40.0 mm · Umean=8 mm/s · η=2.000 mPa·s · T=3.2 s

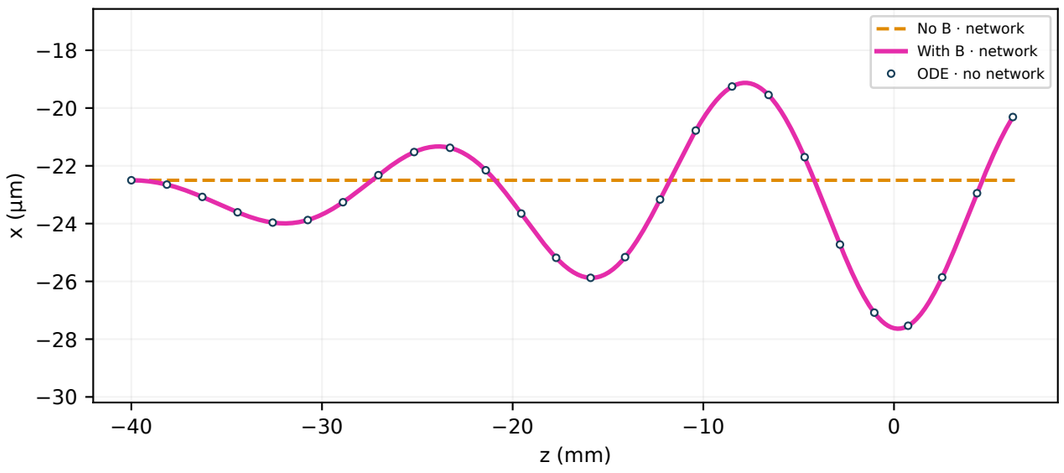
02 Space–time magnetic field



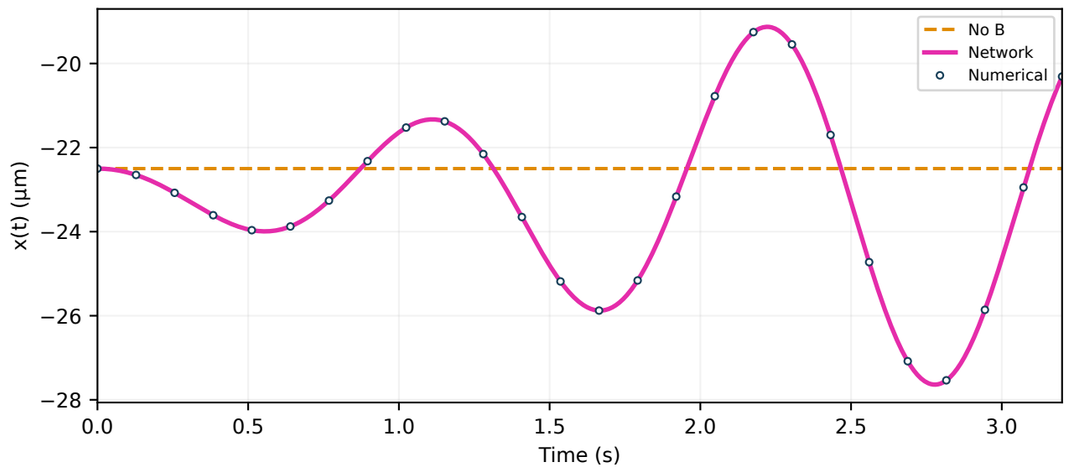
03 Trained-domain fluid flow



04 Network / numerical paths



05 Transverse response



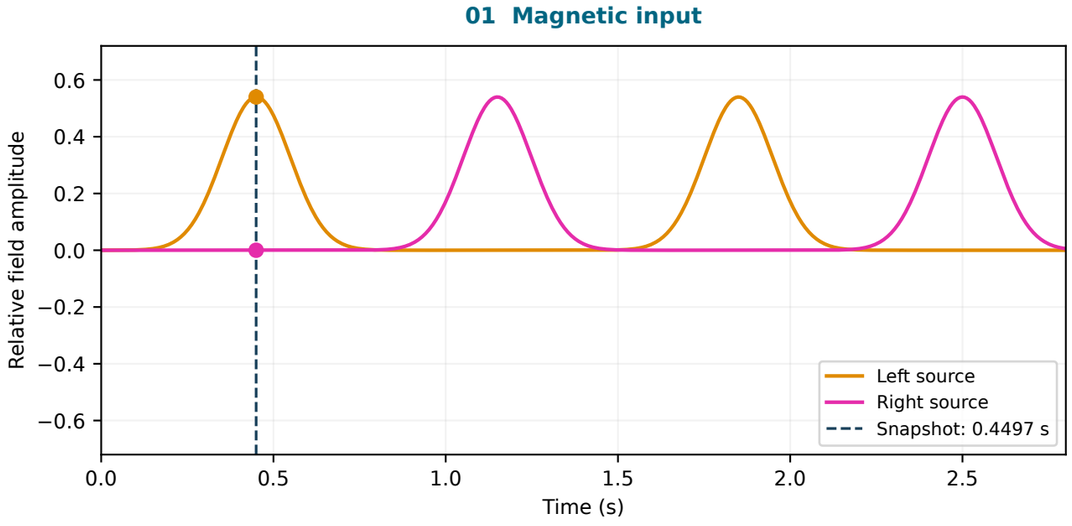
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

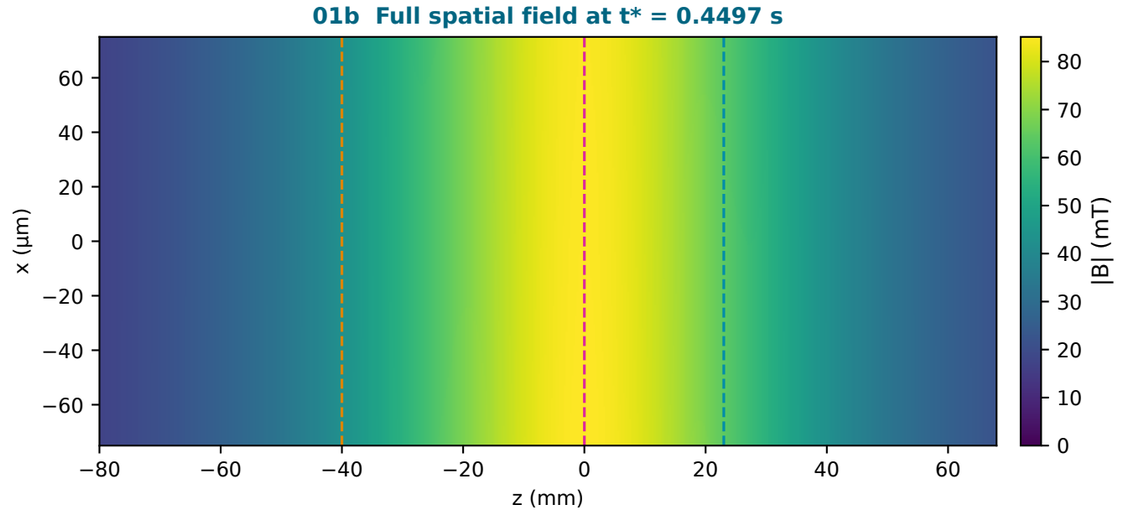
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

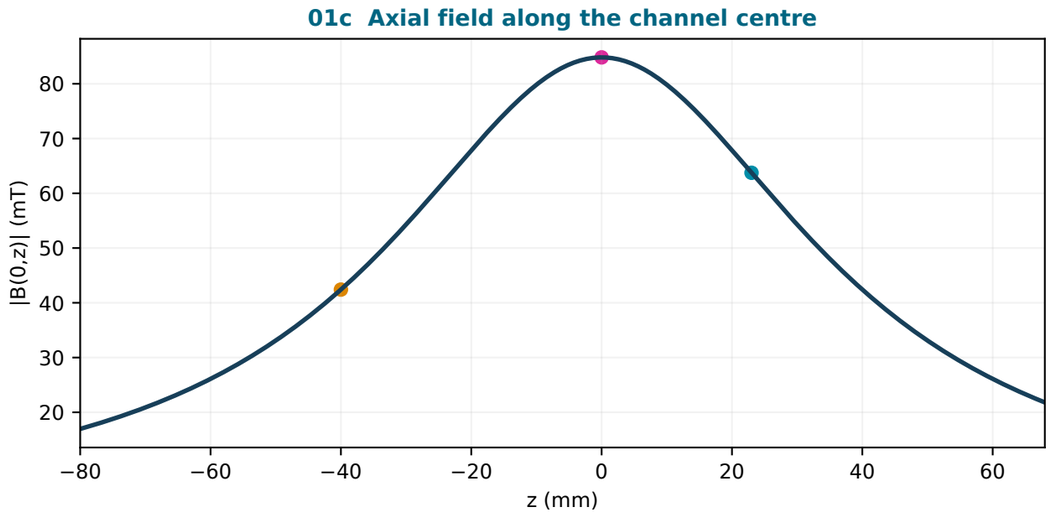
TRAINED MASTER10 | Shifted release · alternating pulse train

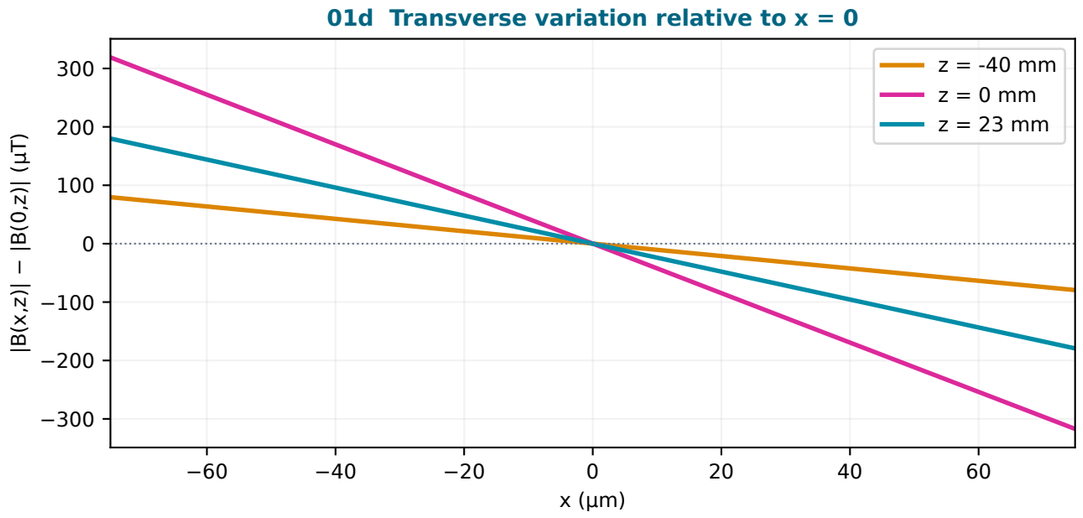
Field detail after input 01 · t*=0.4497 s · a_left=0.5400, a_right=0.0000

Snapshot selection: Peak actual field amplitude







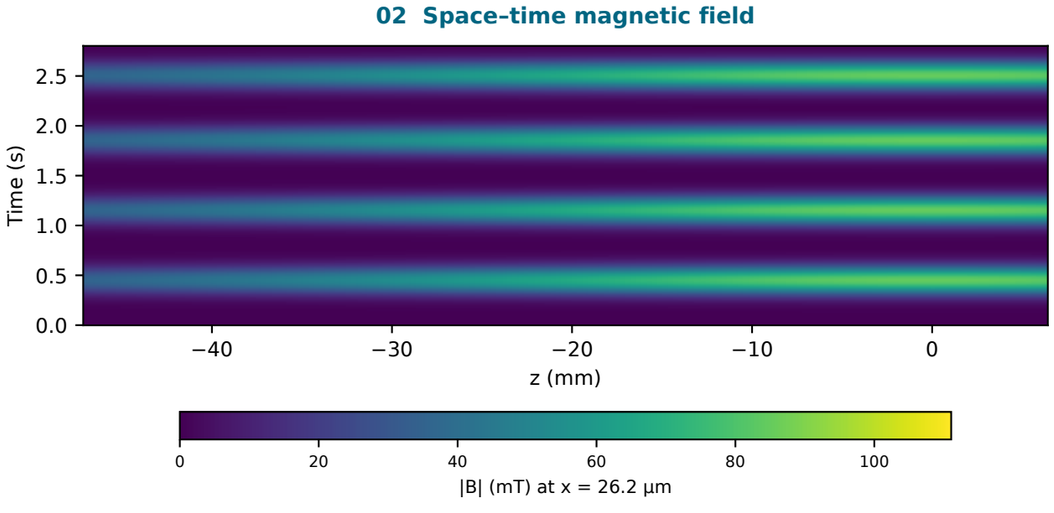


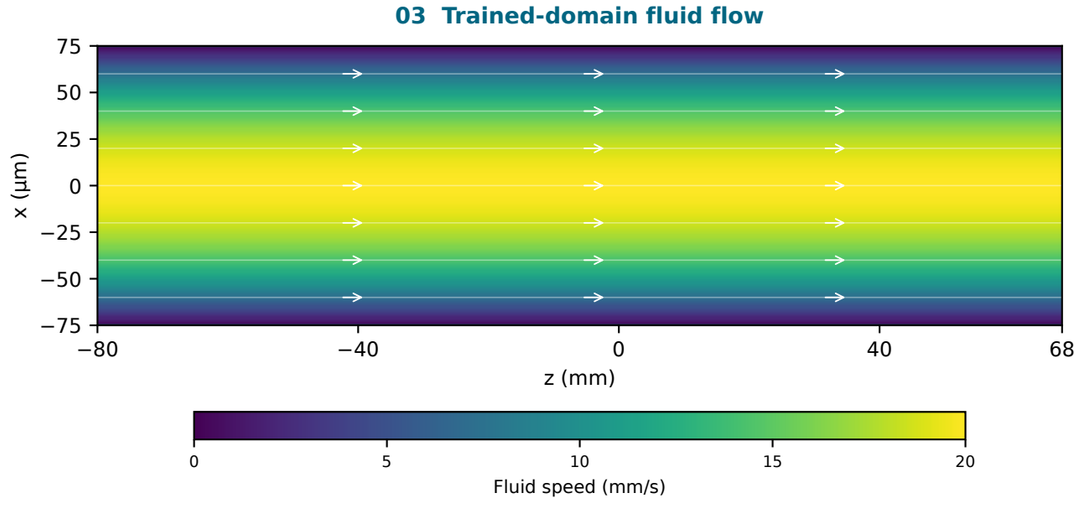
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

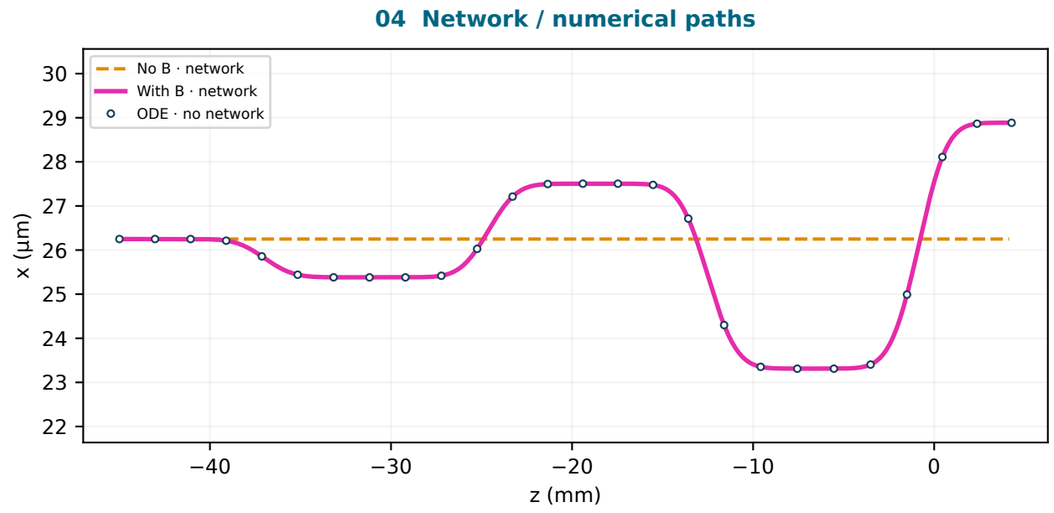
Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

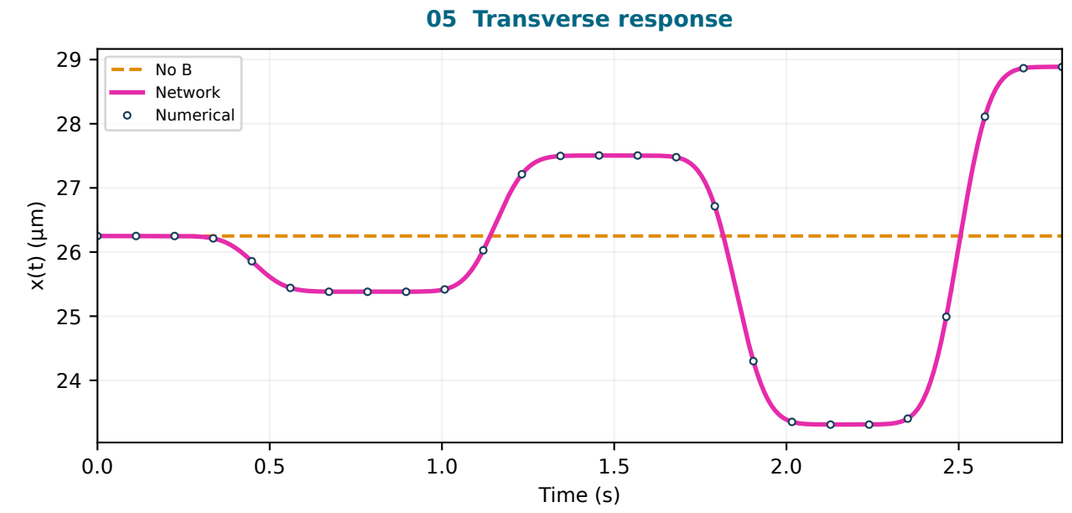
TRAINED MASTER10 | Shifted release · alternating pulse train

Continuation: panels 02–05 · x_0=26.25 µm, z_0=-45.0 mm · Umean=10 mm/s · η=1.500 mPa·s · T=2.8 s









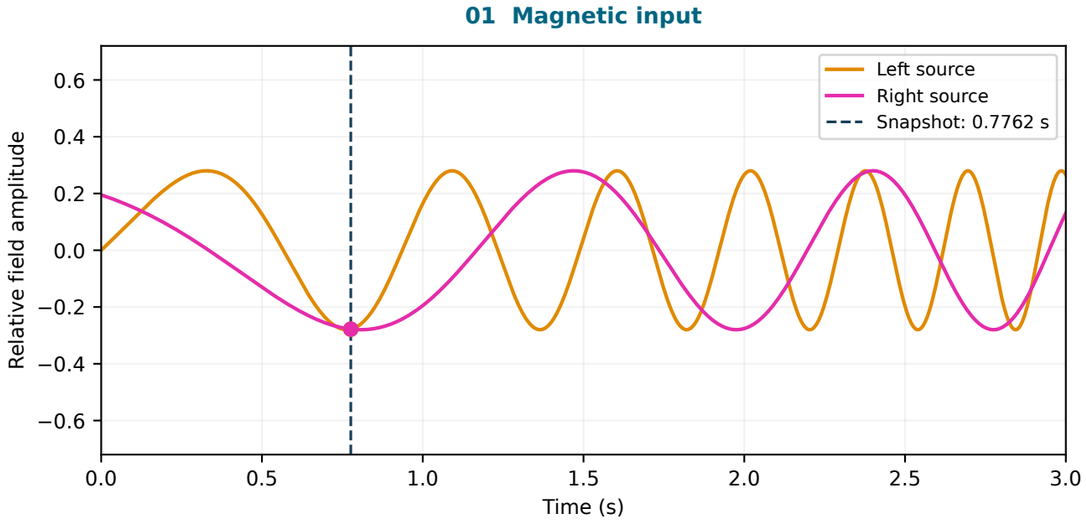
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

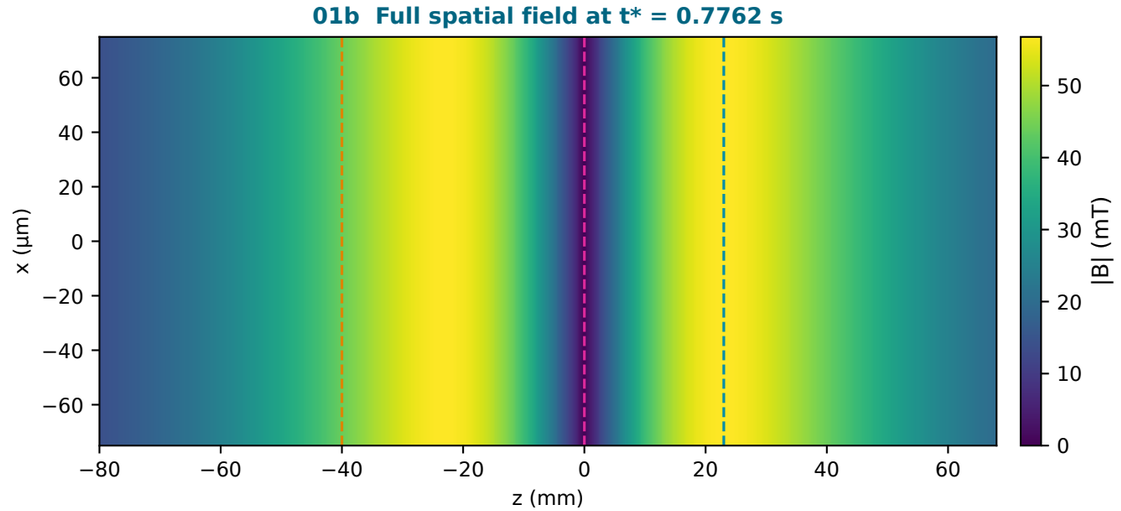
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

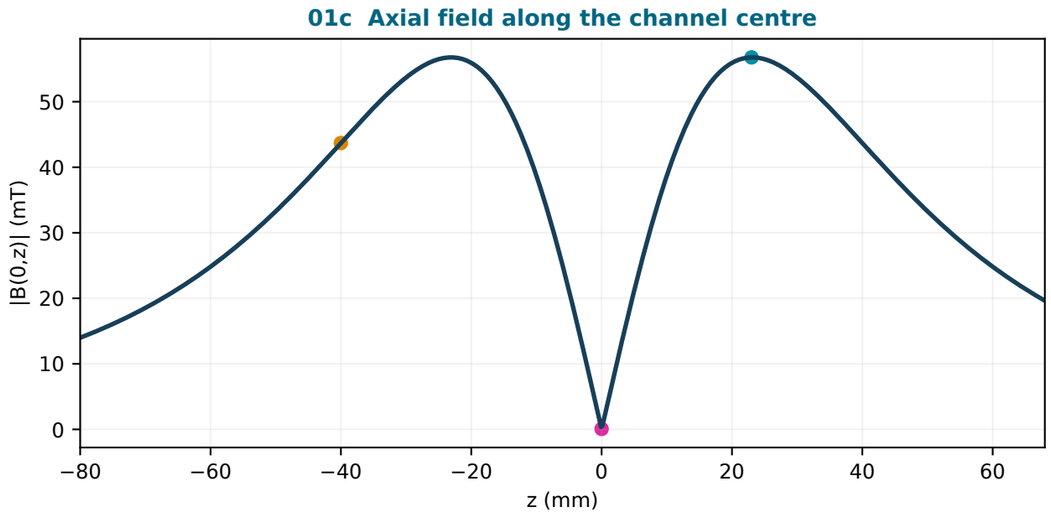
TRAINED MASTER10 | Slow flow · two-source chirps

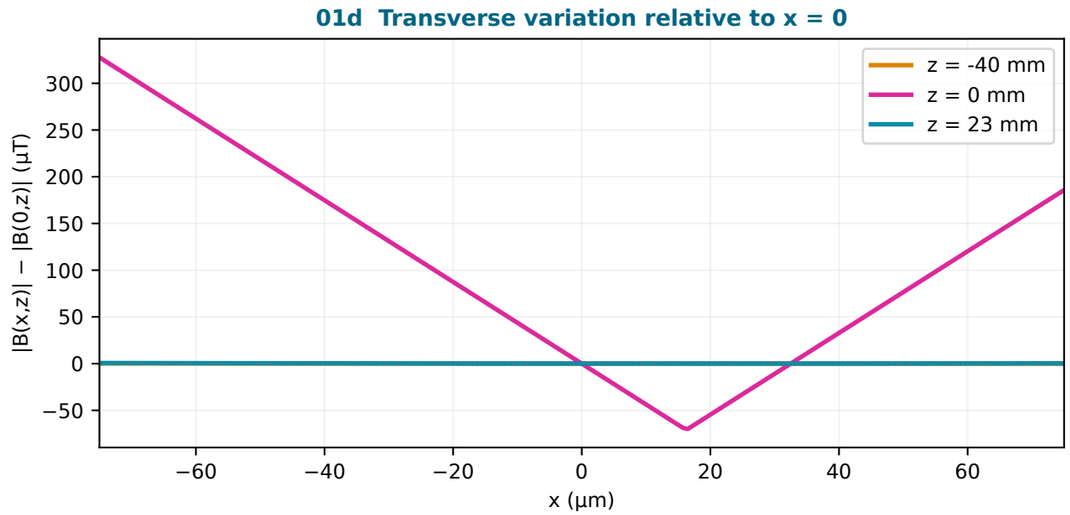
Field detail after input 01 · t*=0.7762 s · a_left=-0.2784, a_right=-0.2779

Snapshot selection: Strong axial interference notch







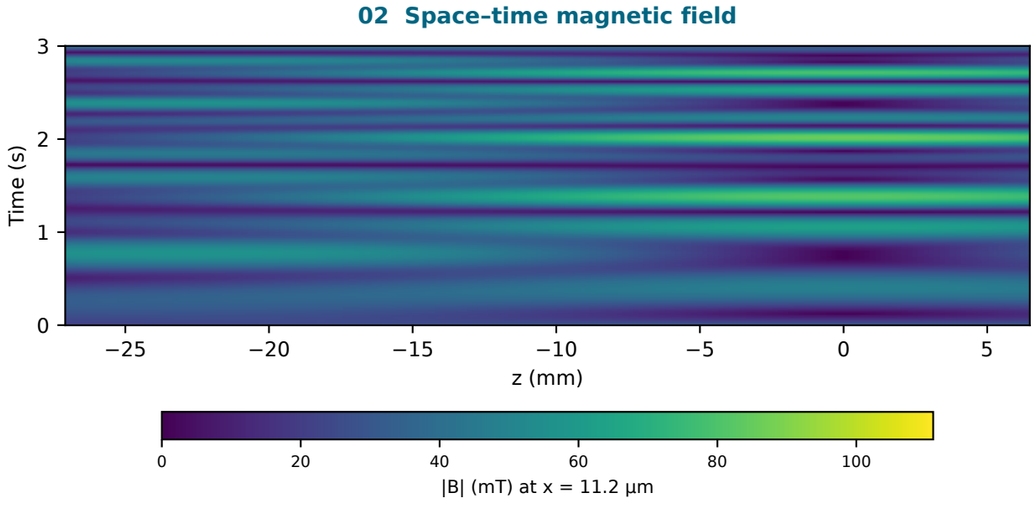


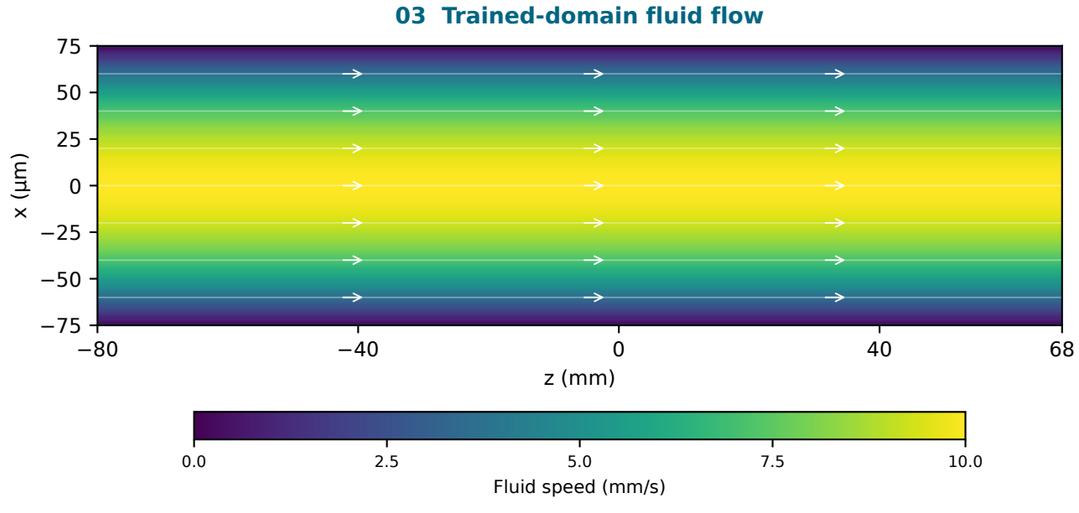
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

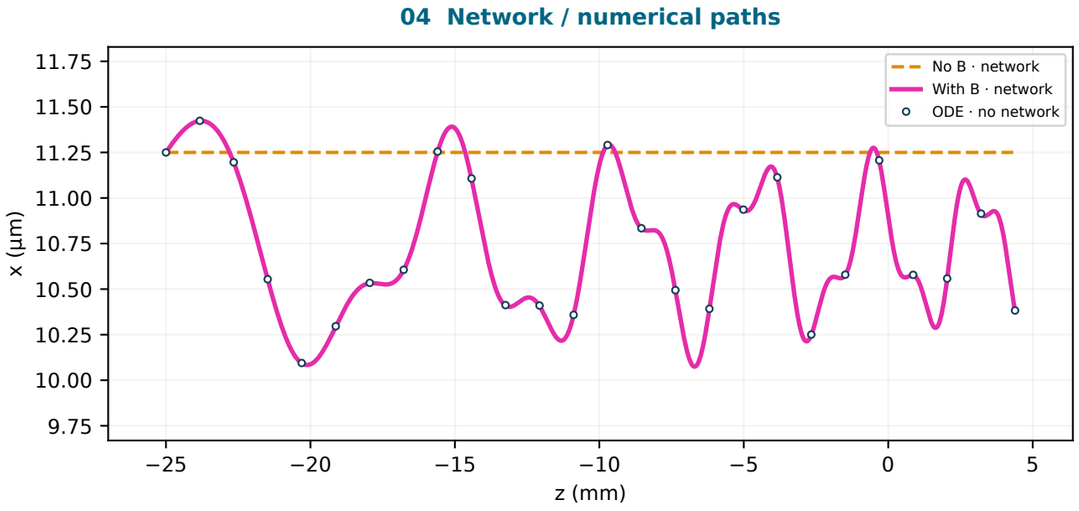
Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

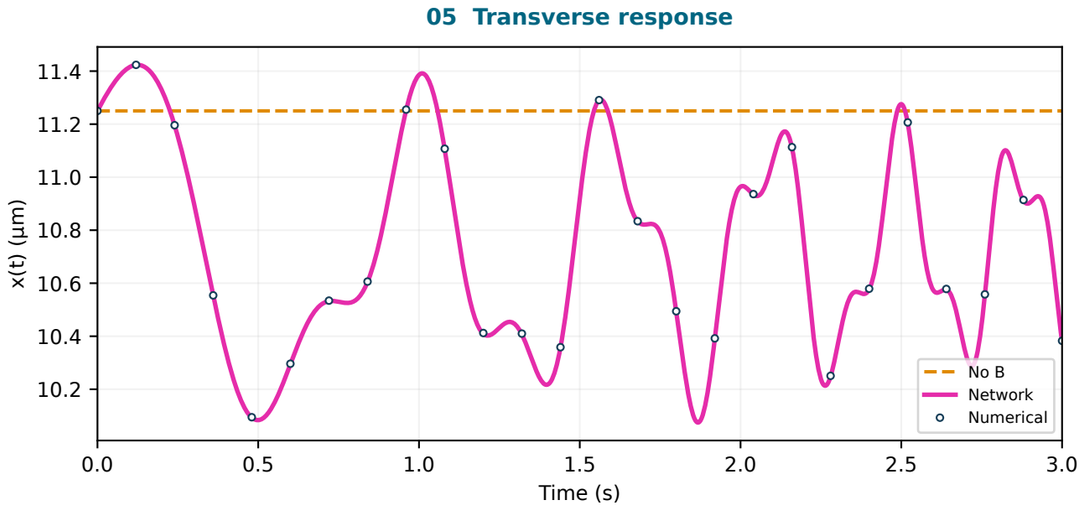
TRAINED MASTER10 | Slow flow · two-source chirps

Continuation: panels 02–05 · x_0=11.25 µm, z_0=-25.0 mm · Umean=5 mm/s · η=1.000 mPa·s · T=3 s









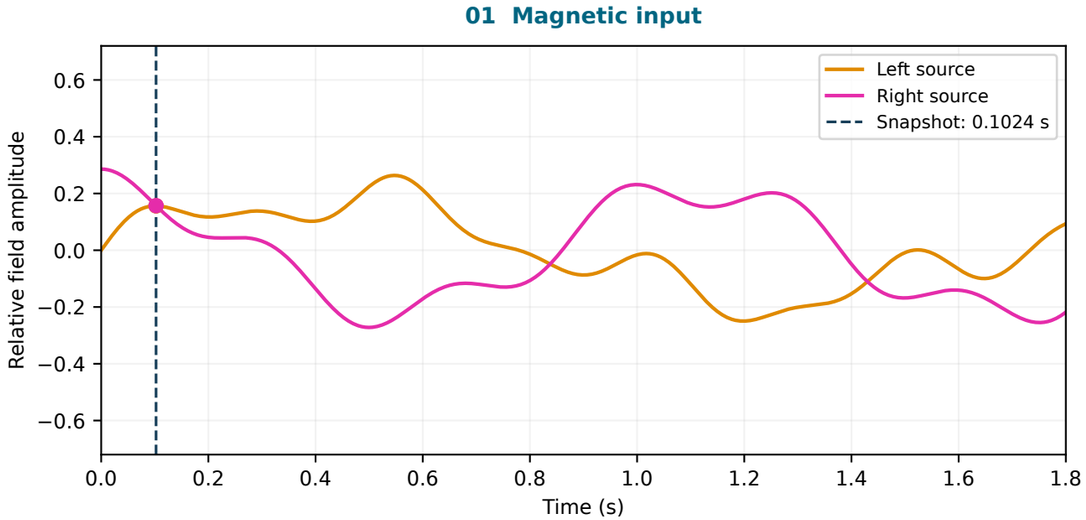
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

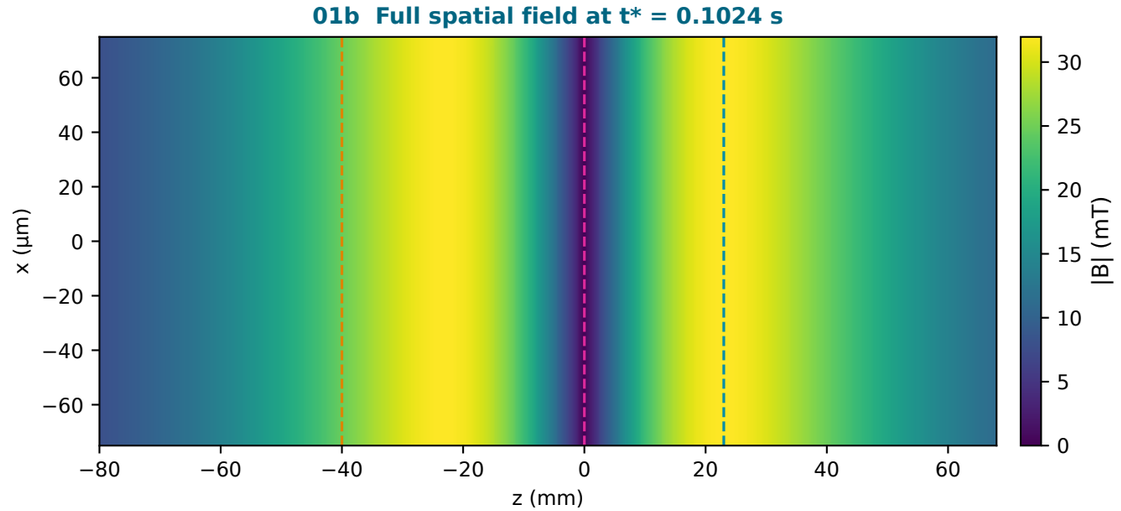
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

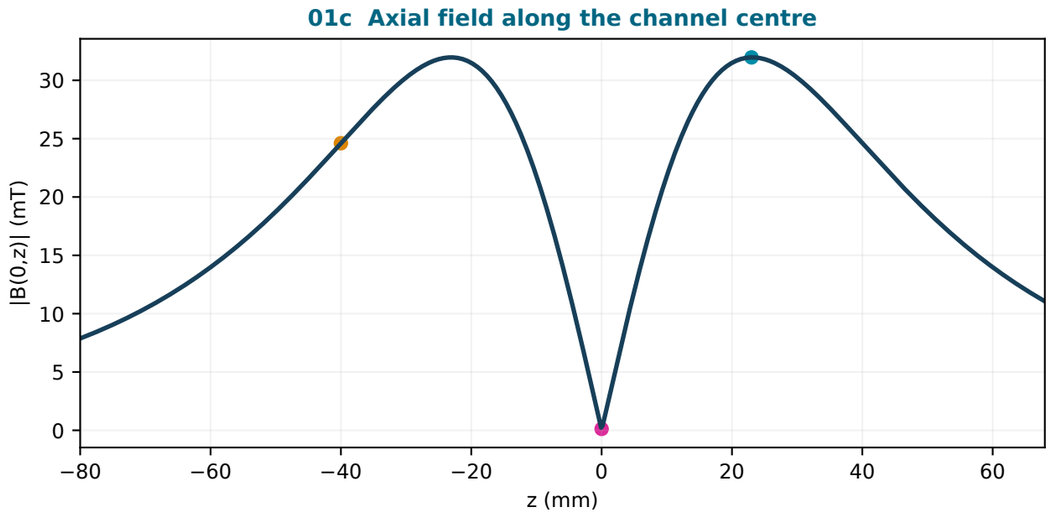
TRAINED MASTER10 | Fast flow · multi-frequency excitation

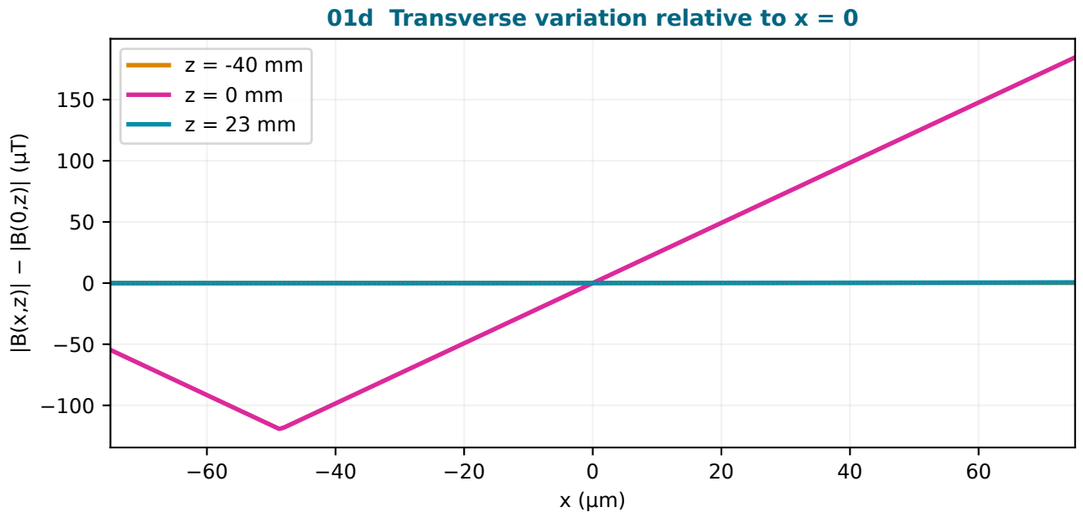
Field detail after input 01 · t*=0.1024 s · a_left=0.1563, a_right=0.1570

Snapshot selection: Strong axial interference notch







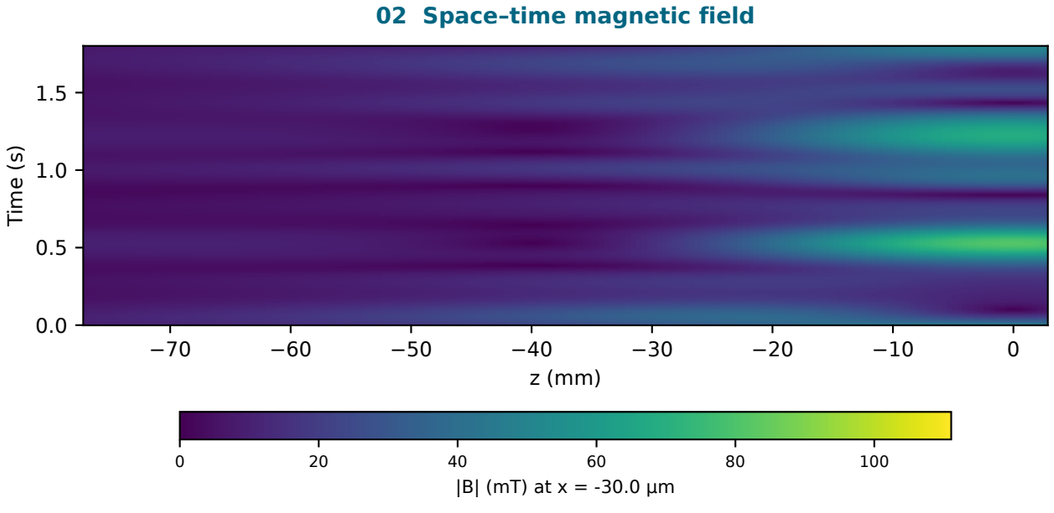


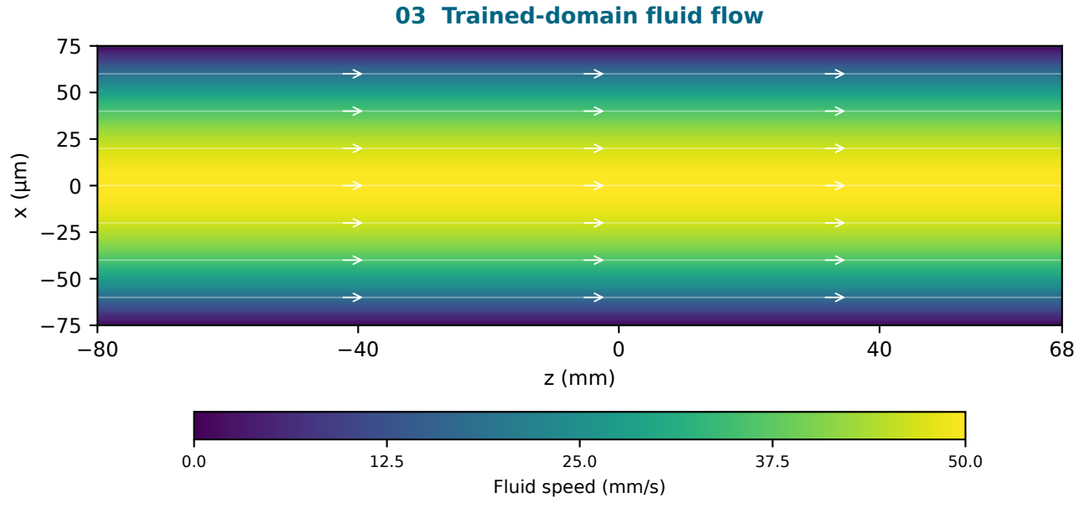
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

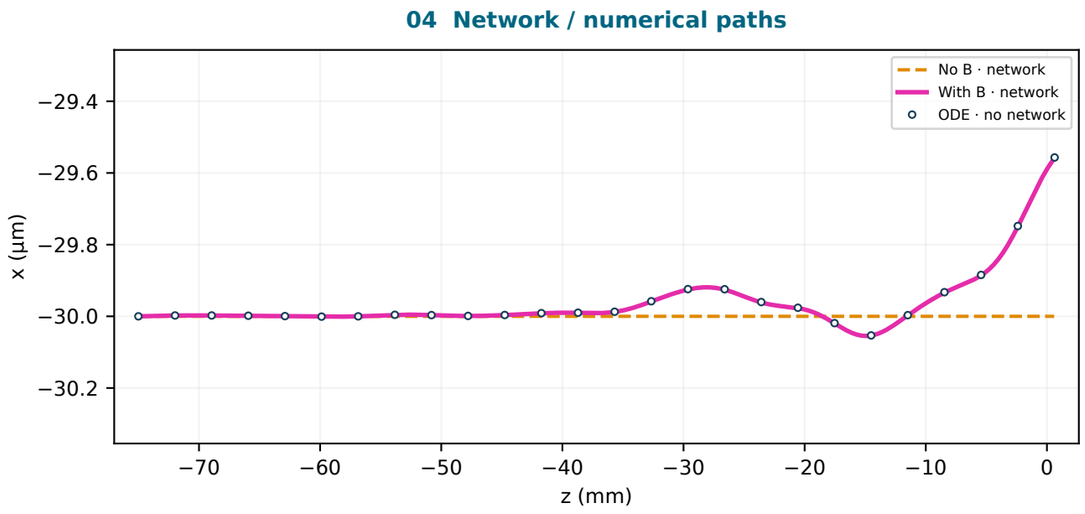
Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

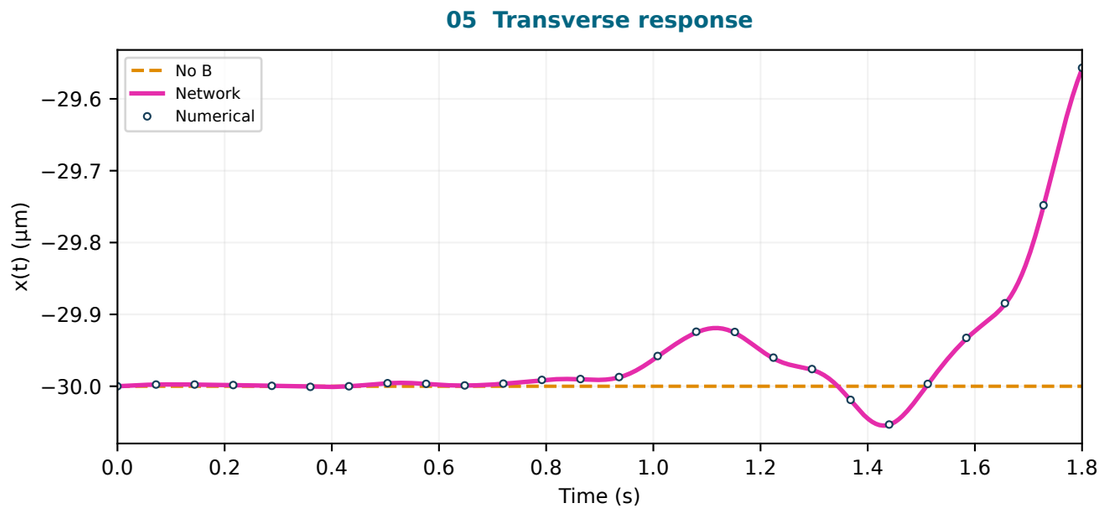
TRAINED MASTER10 | Fast flow · multi-frequency excitation

Continuation: panels 02–05 · x_0=-30.00 µm, z_0=-75.0 mm · Umean=25 mm/s · η=3.500 mPa·s · T=1.8 s









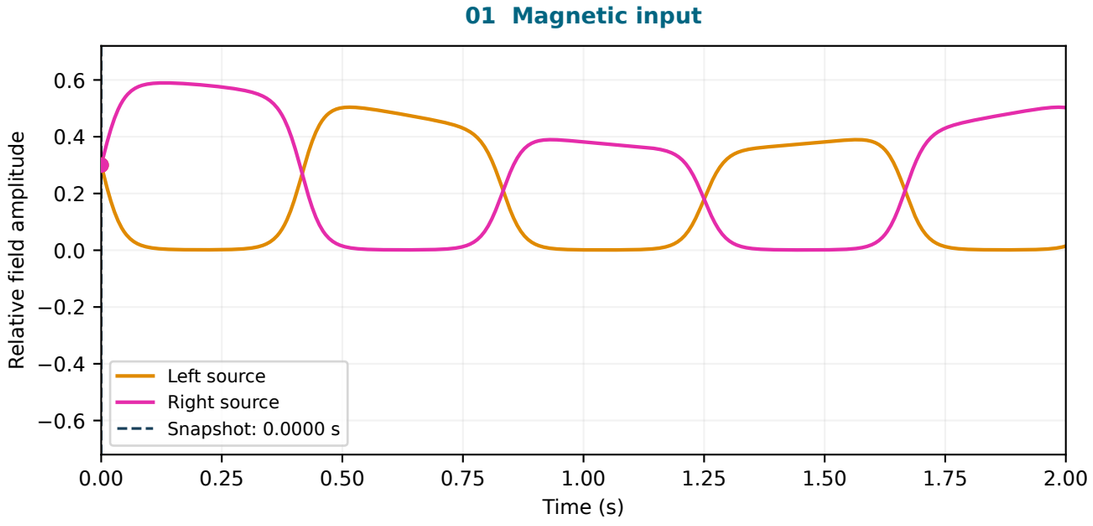
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

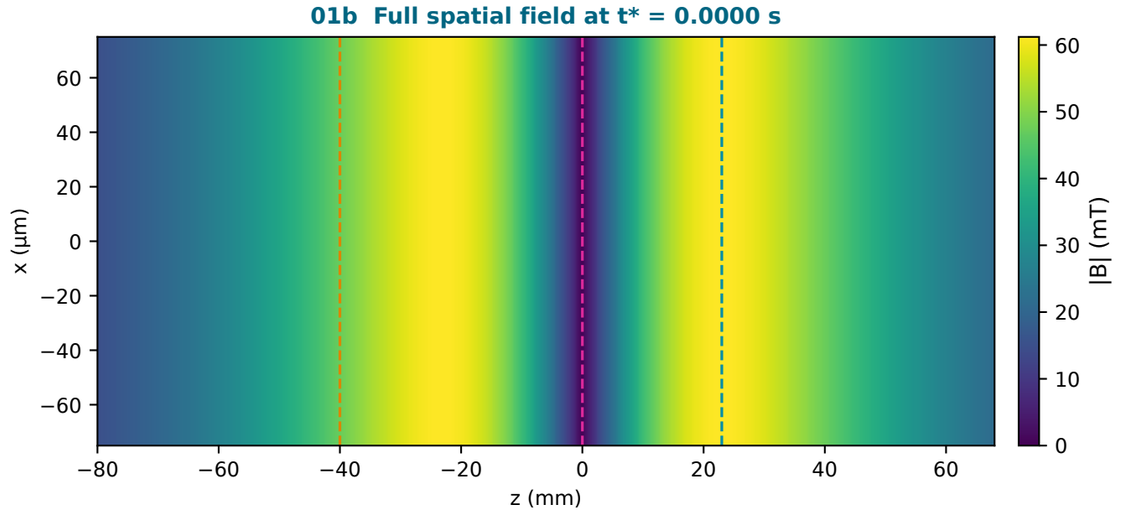
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

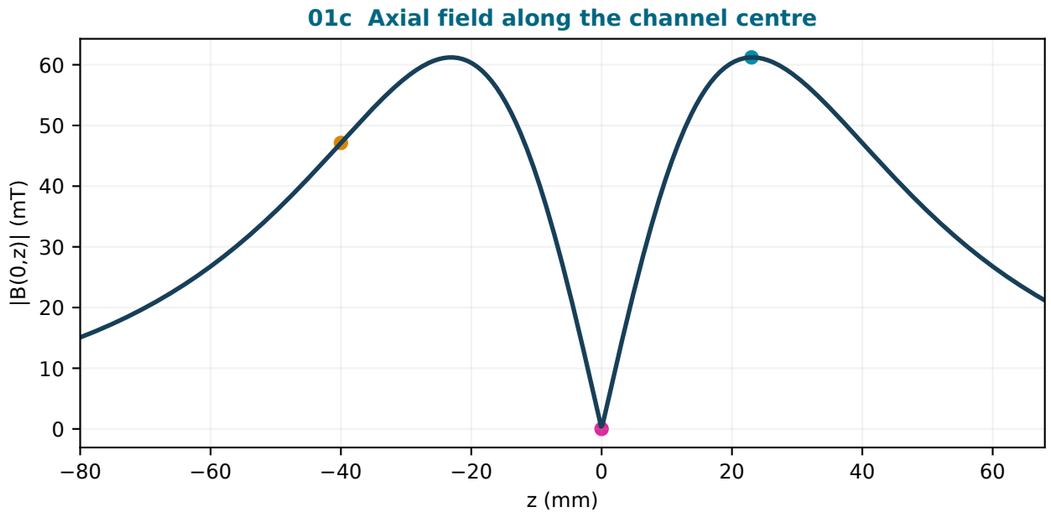
TRAINED MASTER10 | Downstream release · repeated source switching

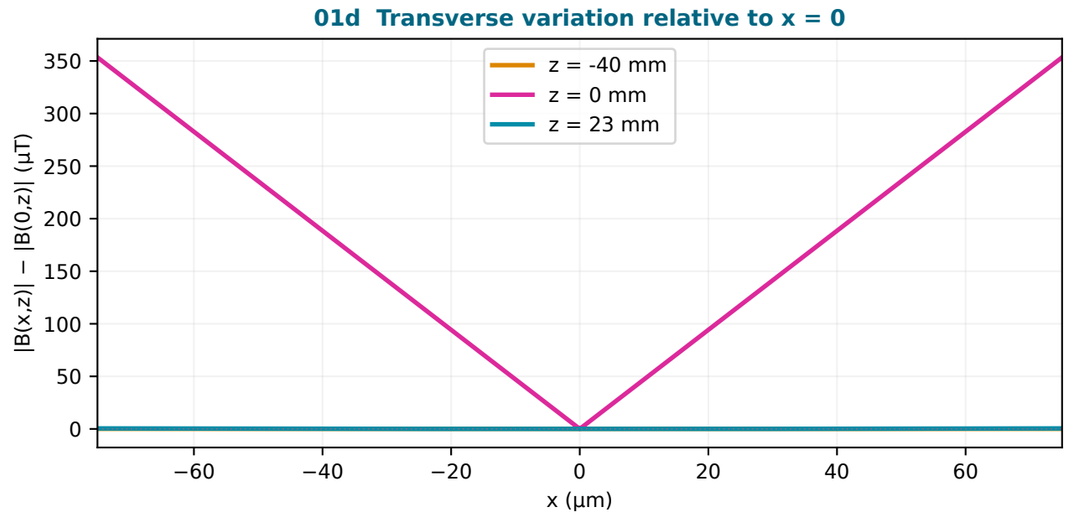
Field detail after input 01 · t*=0.0000 s · a_left=0.3000, a_right=0.3000

Snapshot selection: Strong axial interference notch







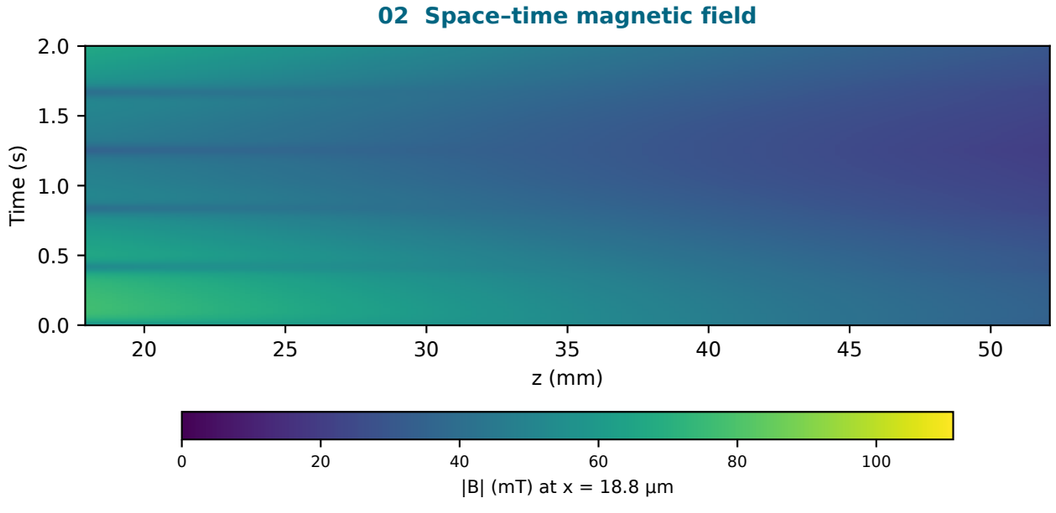


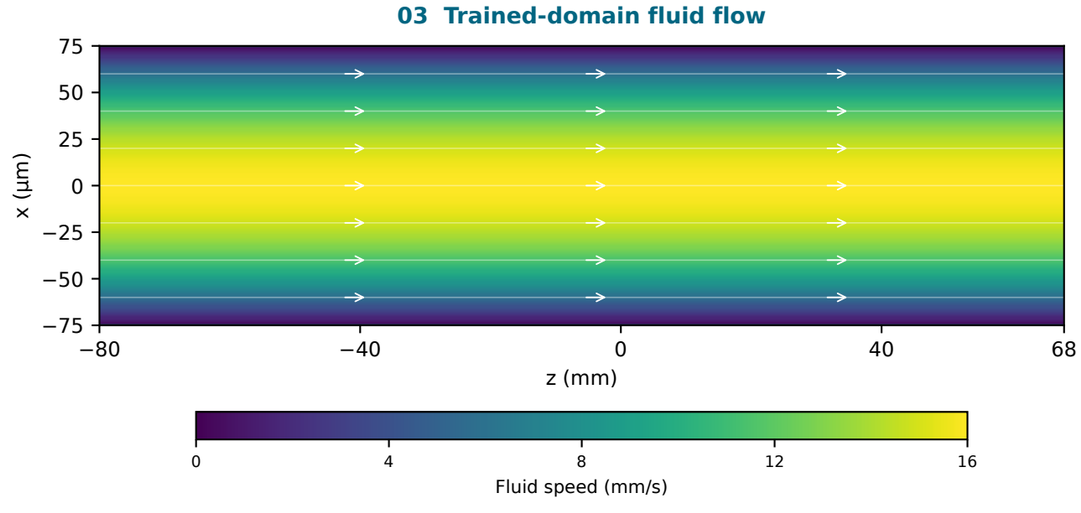
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

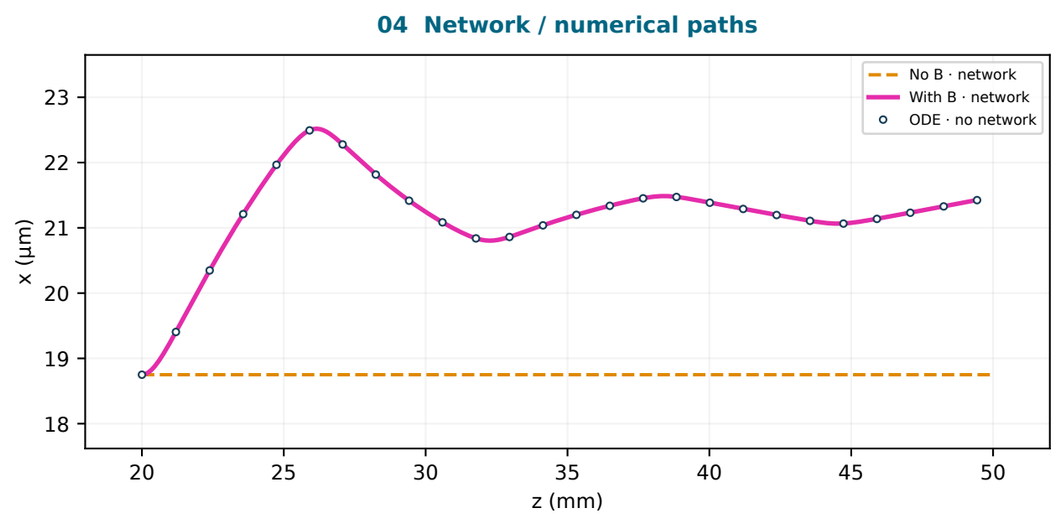
Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

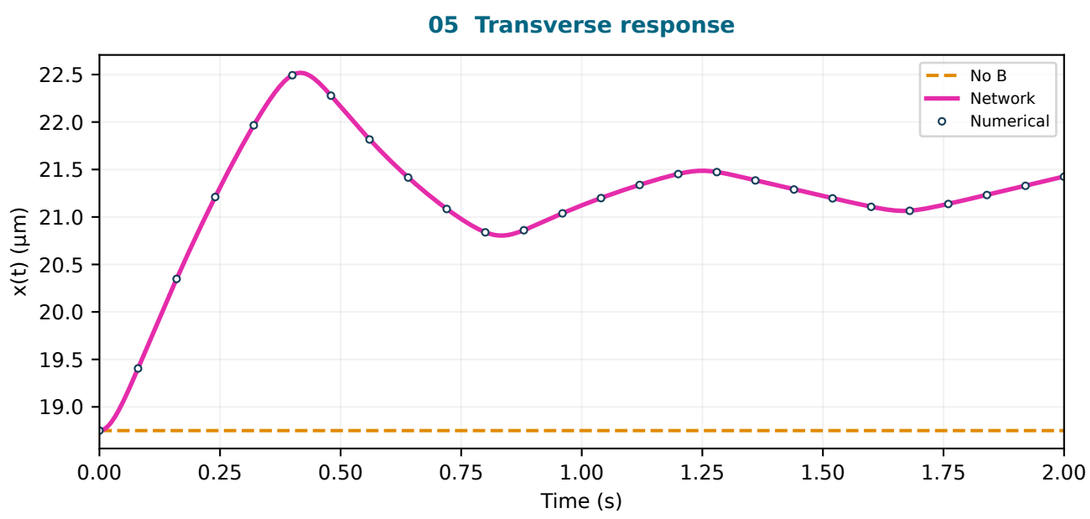
TRAINED MASTER10 | Downstream release · repeated source switching

Continuation: panels 02–05 · x_0=18.75 µm, z_0=20.0 mm · Umean=8 mm/s · η=2.200 mPa·s · T=2 s









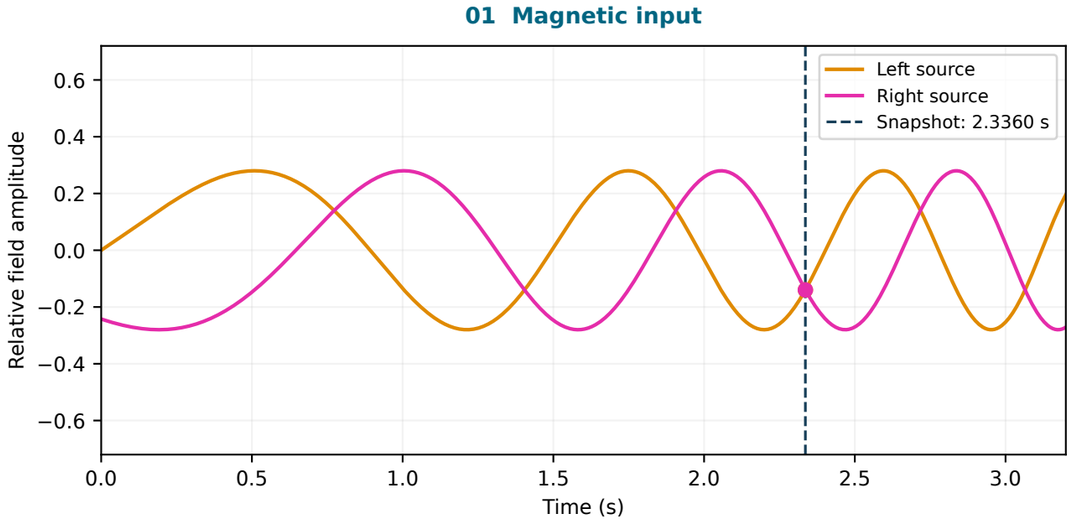
Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

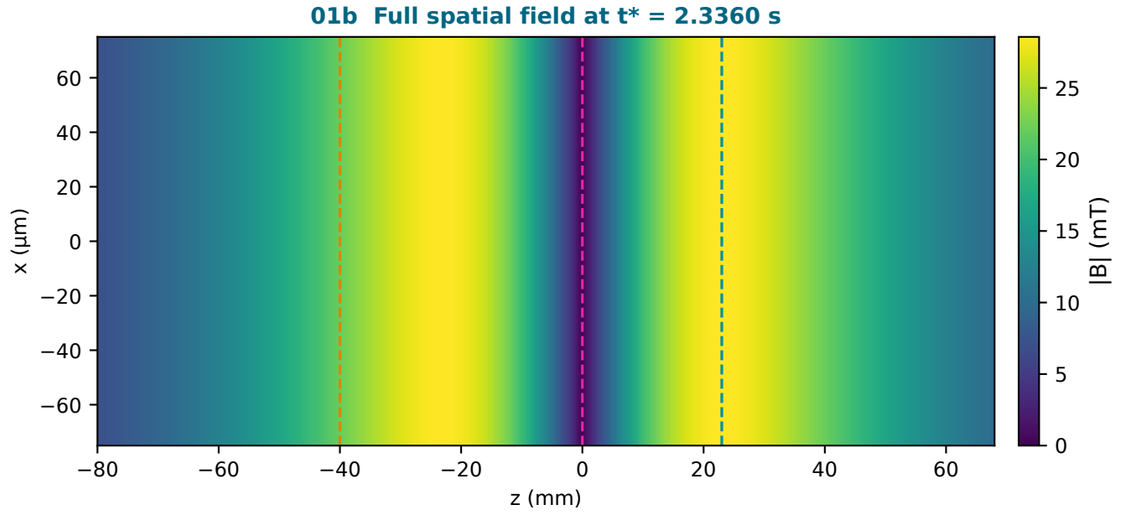
Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.

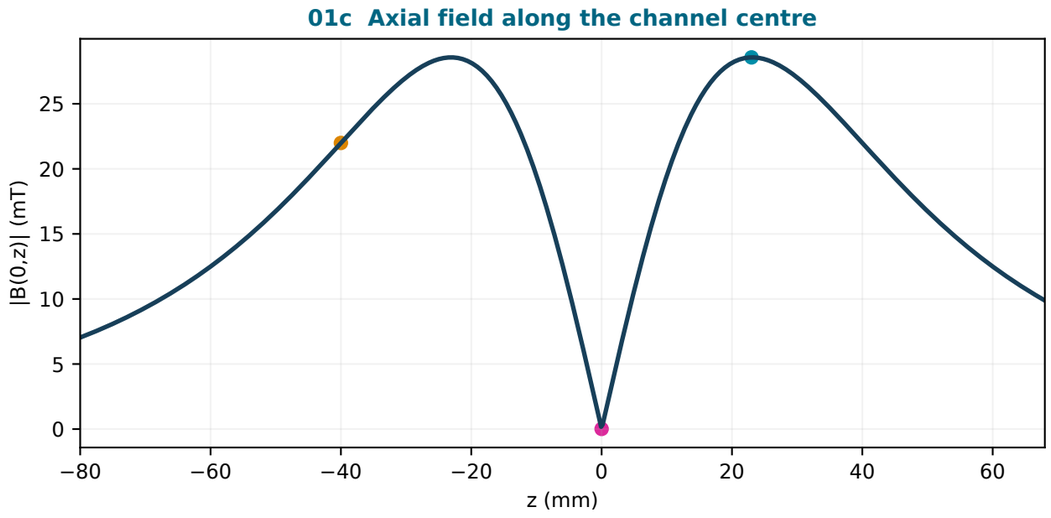
TRAINED MASTER10 | Shifted release · phase-controlled interference

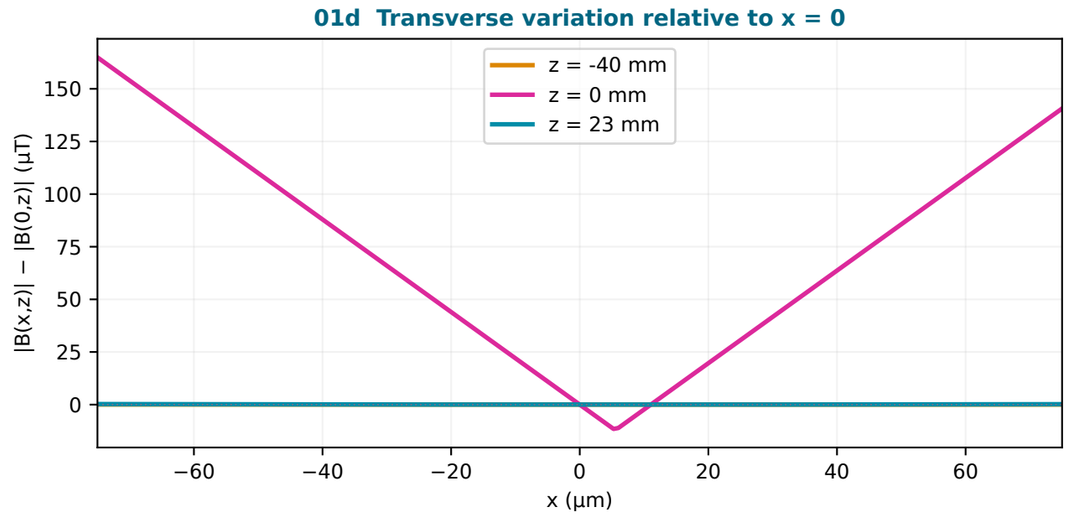
Field detail after input 01 · t*=2.3360 s · a_left=-0.1400, a_right=-0.1400

Snapshot selection: Strong axial interference notch







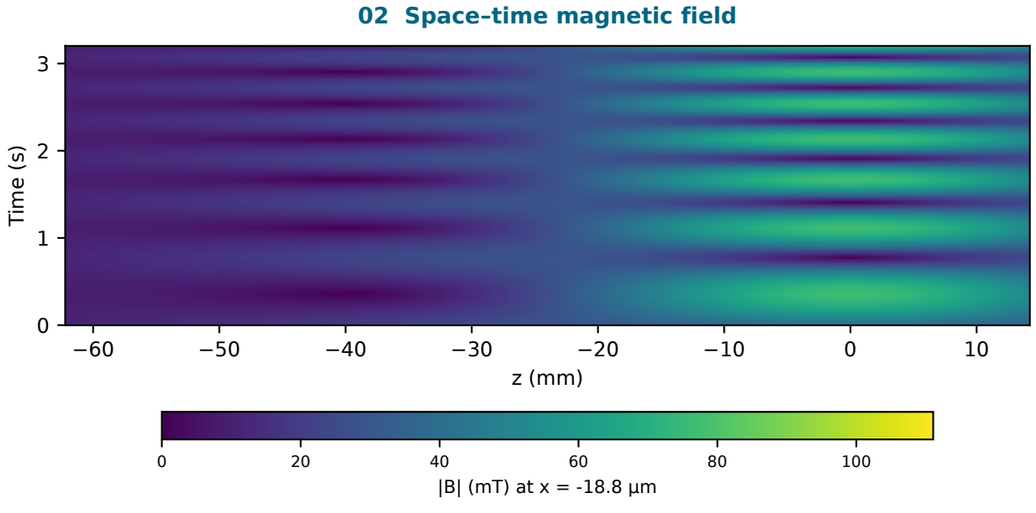


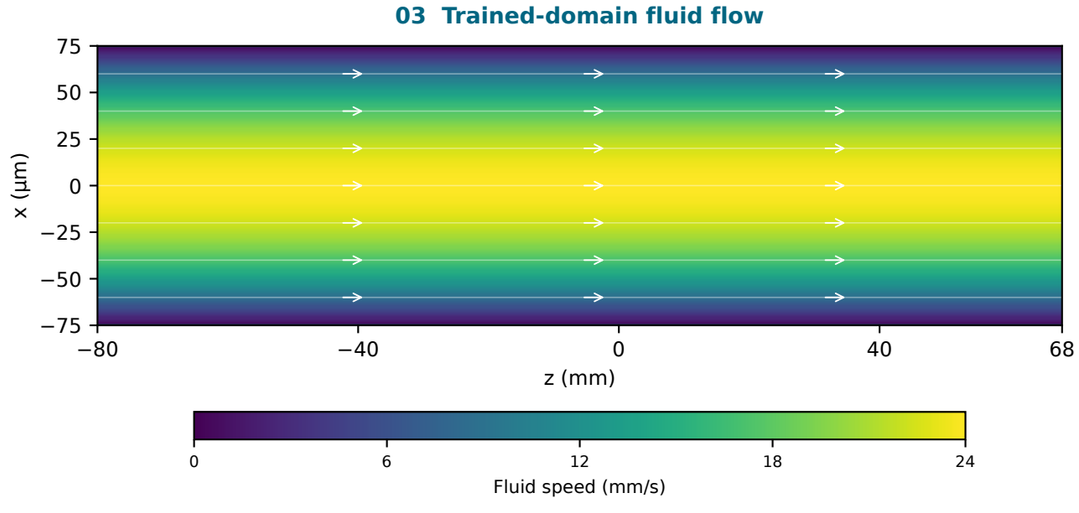
01b/c: actual prescribed field magnitude. 01d: difference from each section centre value, not field magnitude or prediction error.

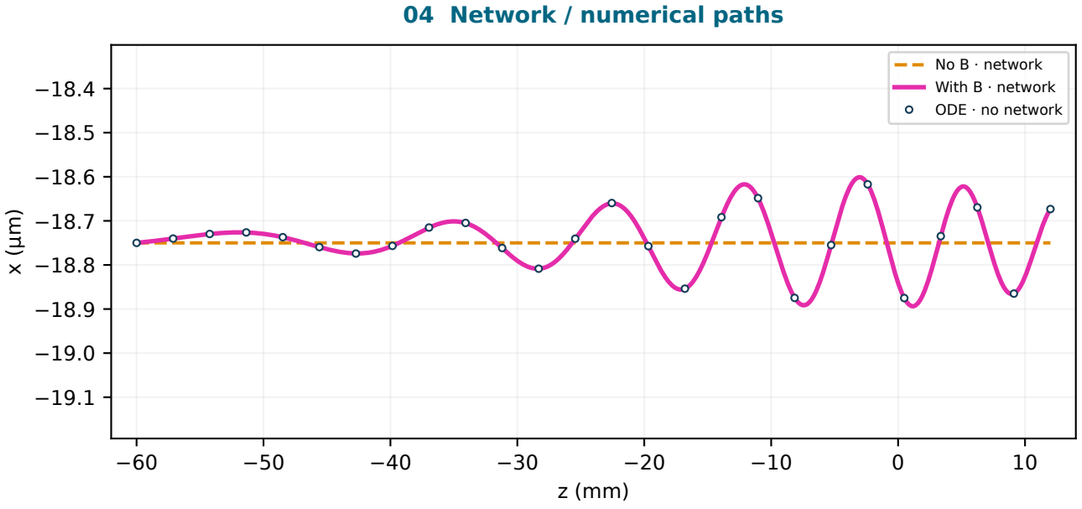
Same two-source geometry and original input functions; no new field learned or invented. Spatial axes use mm and µm.

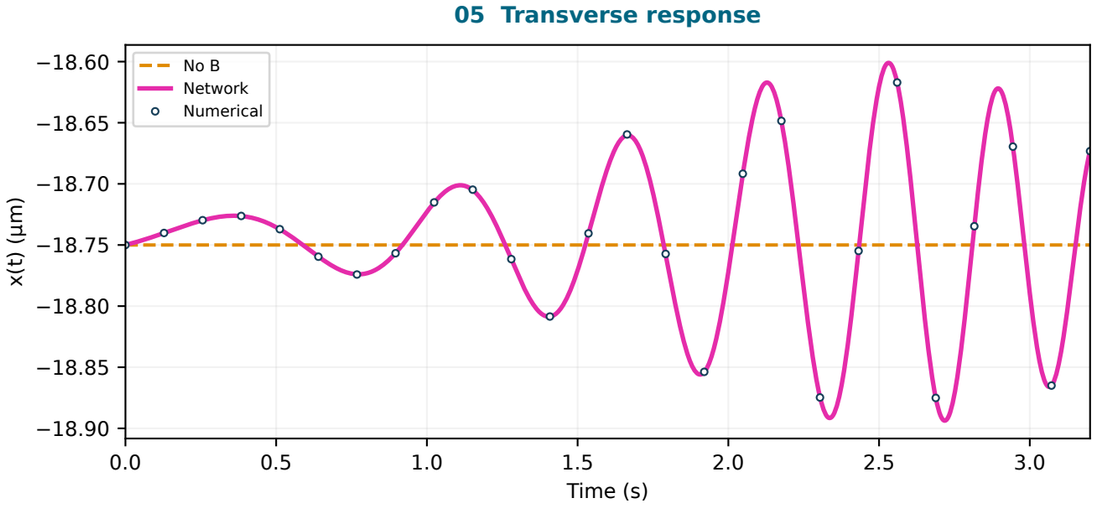
TRAINED MASTER10 | Shifted release · phase-controlled interference

Continuation: panels 02–05 · x_0=-18.75 µm, z_0=-60.0 mm · Umean=12 mm/s · η=4.500 mPa·s · T=3.2 s









Magenta: trained network with B. Dark open circles: direct physical ODE. Orange dashed: same release without B.

Validated trajectories and weights unchanged. Fluid view: full channel; particle paths: zoomed, no fluid-speed colourbar.